* Characteristic Table and Excitation Table:

↳ Truth table contains the inputs and outputs
↳ Excitation Table takes outputs as inputs and Inputs as O/P's. → used for design of counters.
↳ characteristic table has the next state as the output.

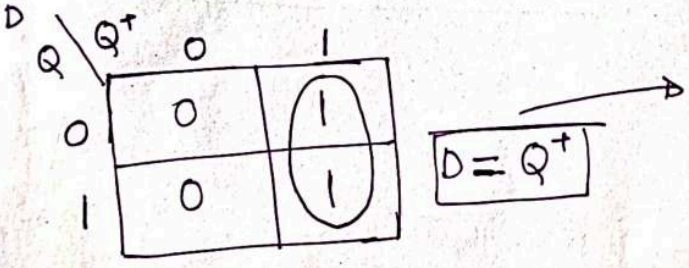
* Truth Table of D-FF

Inputs		outputs	
D	CLK	Q^+	$\bar{Q}^+$
0	↑	0	1
1	↑	1	0
X	0	Q	$\bar{Q}$
X	1	Q	$\bar{Q}$

} No change

Next State Table

D	Q	Q^+
0	0	0
	1	0
1	0	1
	1	1



$D = Q^+$

Excitation Tables:

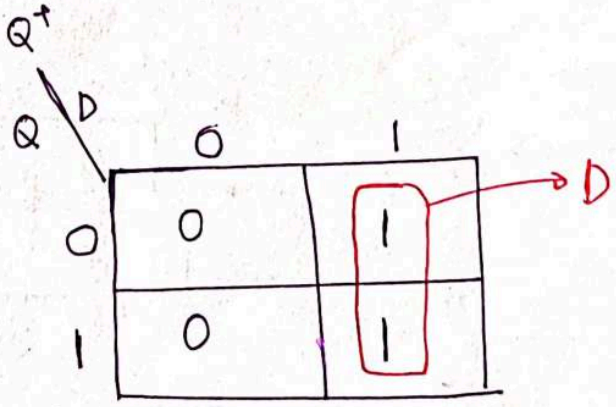
present state	Next State	input D
Q	Q^+	D
0	0	0
0	1	1
1	0	0
1	1	1

Characteristic Table:

present State	input	Next State
Q	D	Q^+
0	0	0
0	1	1
1	0	0
1	1	1

$Q^+ = Q$

D



$Q^+ = D$

characteristic Equation

* JK FF :

Truth Table

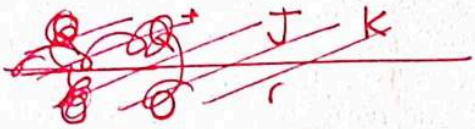
J	k	CLK	Q^+	$\bar{Q}^+$
0	0	↑	Q	$\bar{Q}$
0	1	↑	0	1
1	0	↑	1	0
1	1	↑	$\bar{Q}$	Q

→ No change

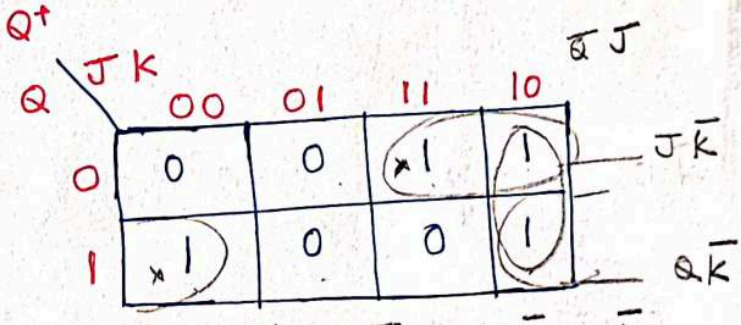
Characteristic Table

present State	Input	Next State
Q	J K	Q^+
0	0 0	0
0	0 1	0
0	1 0	1
0	1 1	1
1	0 0	1
1	0 1	0
1	1 0	1
1	1 1	0

Next State Table:



J	K	Q	Q^+
0	0	0	0
		1	0
0	1	0	0
		1	0
1	0	1	1
		0	1
1	1	1	0
		0	1



$PI's = \bar{Q}J, J\bar{K}, Q\bar{K}$

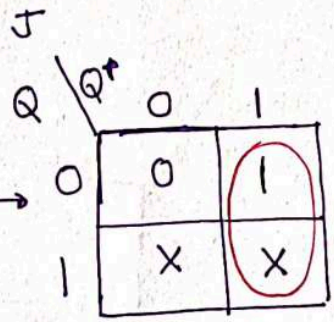
$EPI's = \bar{Q}J, Q\bar{K}$

$\therefore Q^+ = \bar{Q}J + Q\bar{K}$ → Characteristic Equation

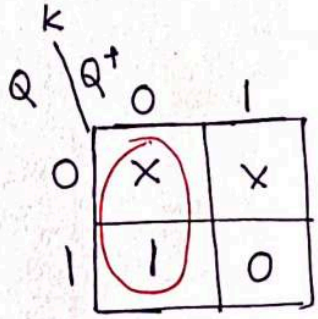
Excitation Table

PS	NS	I/p's	
Q	Q^+	J	$\bar{K}$
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

→



$J = Q^+$



$K = \bar{Q}^+$

* SR FF

Truth Table

S	R	CLK	Q^+	$\bar{Q}^+$	
0	0	⊓	Q	$\bar{Q}$	No change
0	1	⊓	0	1	
1	0	⊓	1	0	
1	1	⊓	1*	1* (Illegal)	Invalid
X	X	0	Q	$\bar{Q}$	No change

Next State Table:

S	R	Q	Q^+
0	0	0	0
		1	1
0	1	0	0
		1	0
1	0	0	1
		1	1
1	1	X	X

Excitation Table

PS	NS	I/P's	
Q	Q^+	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

S

Q \ Q^+	0	1
0	0	1
1	0	X

$S = Q^+$

R

Q \ Q^+	0	1
0	X	0
1	1	0

$R = \bar{Q}^+$

Characteristic Table:

present State	Input		Next State
Q	S	R	Q^+
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X

Q^+

Q \ SR	00	01	11	10
0	0	0	X	1
1	1	0	X	1

$Q\bar{R}$ S

$$Q^+ = S + Q\bar{R}$$

↳ characteristic Equation

* T-FF

Truth Table

Inputs		outputs	
T	CLK	Q^+	$\bar{Q}^+$
1	↑	$\bar{Q}$	Q
0	↑	Q	$\bar{Q}$
1	0	Q	$\bar{Q}$
1	1	Q	$\bar{Q}$

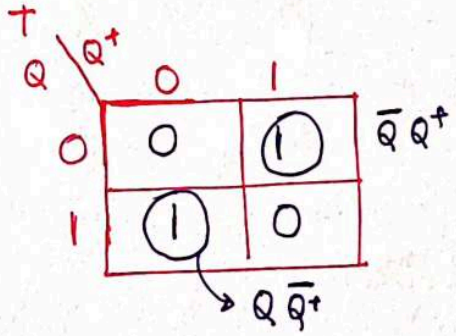
} NC

Next state Table

T	Q	Q^+
1	0	1
	1	0
0	0	0
	1	1

Excitation Table:

Q	Q^+	T
0	0	0
0	1	1
1	0	1
1	1	0

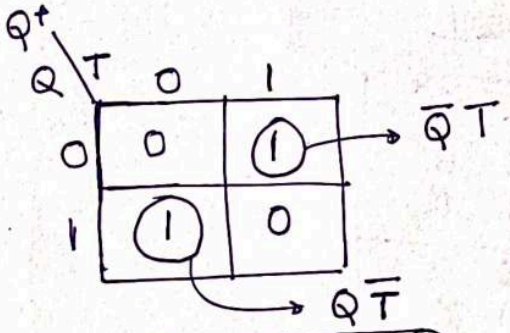


$\therefore T = Q^+\bar{Q} + \bar{Q}^+Q$

$T = Q \oplus Q^+$

characteristic Table

present state	Input	Next state
Q	T	Q^+
0	0	0
0	1	1
1	0	1
1	1	0



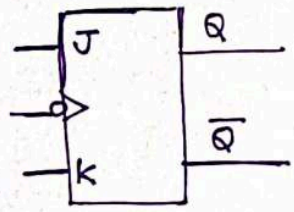
$\therefore Q^+ = Q\bar{T} + \bar{Q}T$ → Characteristic Equation

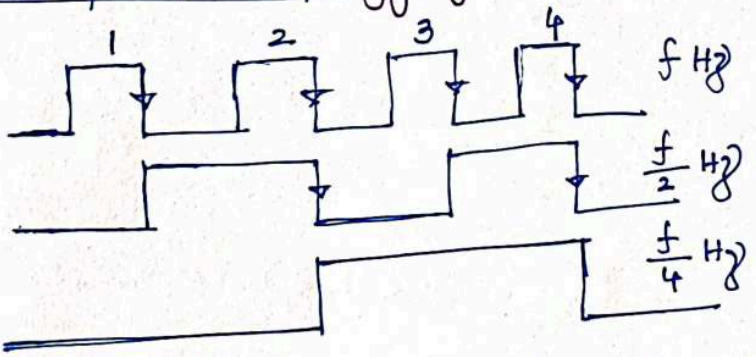
$Q^+ = Q \oplus T$

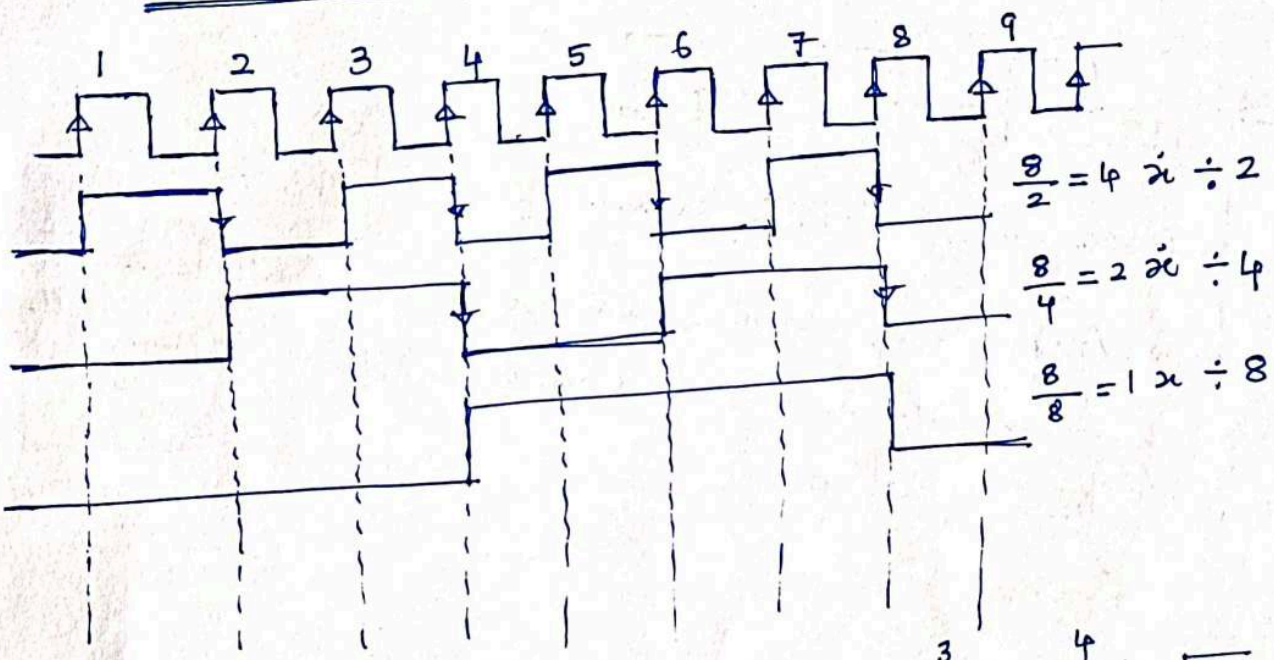
* Frequency Divider circuit :

1)

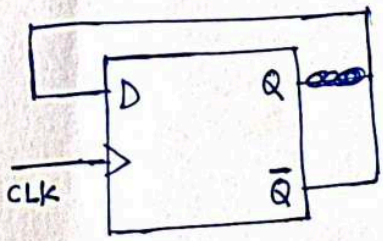
J	k	Q^+	$\bar{Q}^+$	
0	0	Q	$\bar{Q}$	No change
0	1	0	1	
1	0	1	0	
1	1	$\bar{Q}$	Q	Toggling

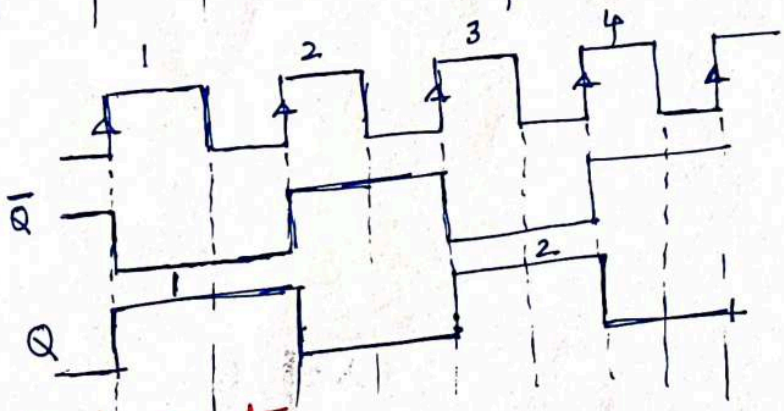






2)





We have, $Q^+ = D$ → characteristic Equation

Given $D = \bar{Q}$

Then $Q^+ = \bar{Q}$ → For each clock transition, the output is complemented each time i.e it becomes from 0 to 1 and 1 to 0 for each clock transition

* Summary:

1) $Q^+ = \bar{Q}J + Q\bar{K}$; $J = Q^+$, $K = \bar{Q^+}$

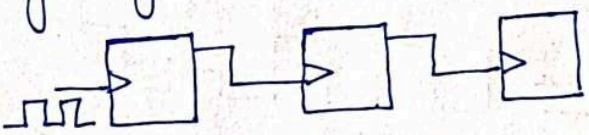
2) $Q^+ = S + Q\bar{R}$; $S = Q^+$, $R = \bar{Q^+}$

3) $Q^+ = D$; $D = Q^+$

4) $Q^+ = Q \oplus T$; $T = Q \oplus Q^+$

Q) Why Asynchronous counter is called Ripple counter?

Ans:



↳ only the first FF is clocked by an external clock

↳ Asynchronous counters are also called as Ripple counters because of the way the clock pulse ripples it way through the flip-flops.

* Flip flop conversion :

NOTE:
SR, JK, T, D FF's.

1) JK to T-FF :

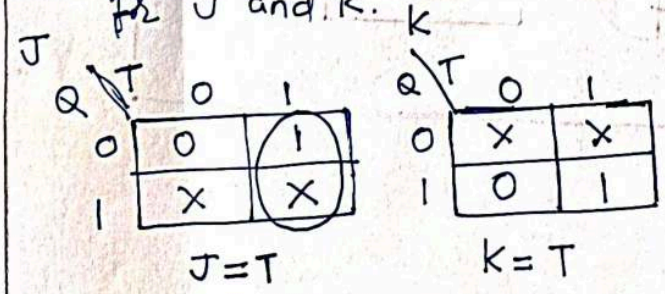
→ Excitation Table of JK

Q	Q^+	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

→ characteristic Table of T-FF

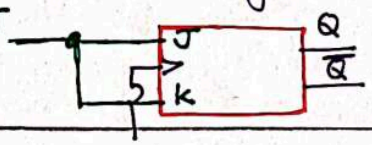
Q	T	Q^+	J	K
0	0	0	0	x
0	1	1	1	x
1	0	ϕ	x	0
1	1	0	x	1

→ obtain the Boolean expression for J and K.



∴ J=K=T

→ Draw the circuit diagram



2) JK to D-FF

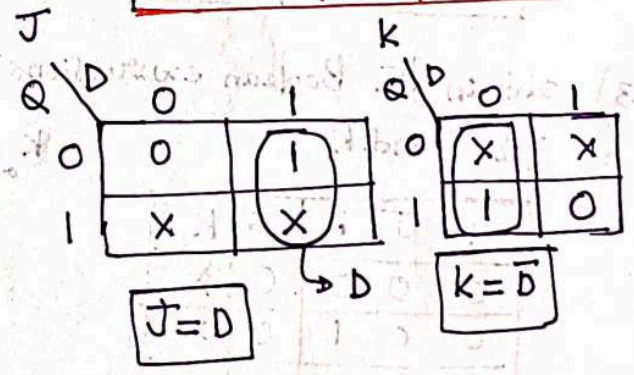
→ Excitation Table of JK FF

→ characteristic Table of D-FF

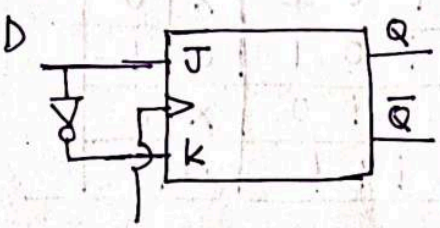
Q	D	Q^+
0	0	0
0	1	ϕ
1	0	0
1	1	1

→ obtain the Boolean expression for J and K.

Q	D	Q^+	J	K
0	0	0	0	x
0	1	1	1	x
1	0	0	x	1
1	1	1	x	0



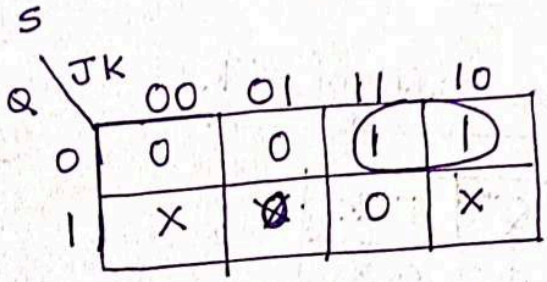
* Draw the circuit diagram



3) SR FF to JK FF Conversion :

1) Excitation Table of SR FF

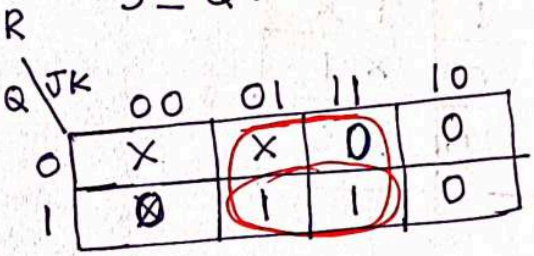
Q	Q^+	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0



$S = \bar{Q}J$

2) Characteristic Table of JK

Q	J	K	Q^+
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

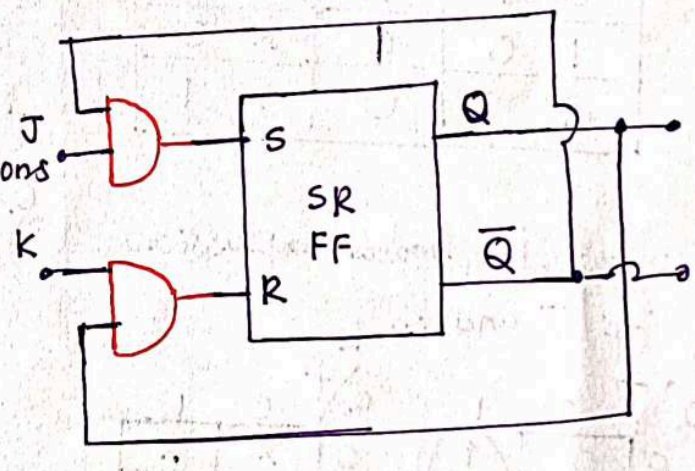


$R = KQ$

3) Obtain the Boolean expressions for S and R

Q	J	K	S	R
0	0	0	0	X
0	0	1	0	X
0	1	0	1	0
0	1	1	1	0
1	0	0	X	0
1	0	1	0	1
1	1	0	X	0
1	1	1	0	1

* circuit diagram :



* D to T FF Conversion :

→ Excitation Table of D-FF

Q	Q^+	D
0	0	0
0	1	1
1	0	0
1	1	1

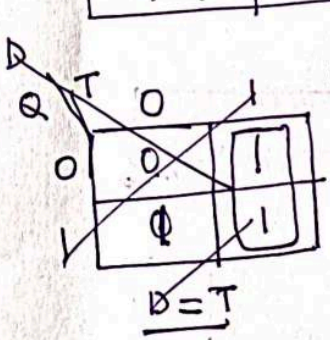
→ Characteristic Table of T FF

Q	T	Q^+	D
0	0	0	0
0	1	1	1
1	0	1	1
1	1	Q	$\bar{Q}$

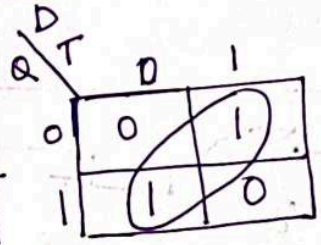
0 0 — 0
0 1 — 1
1 1 — 1
1 0 — 0

→ obtain the expression for D

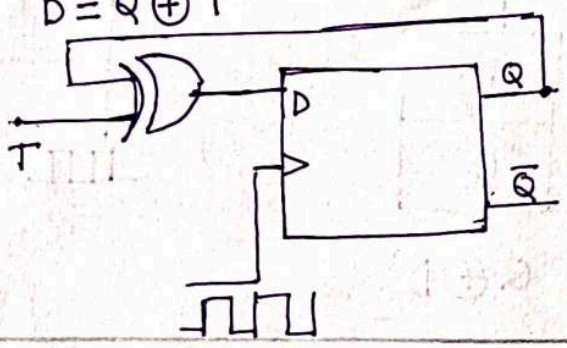
Q	T	D
0	0	0
0	1	1
1	0	1
1	1	0



D = T



D = Q ⊕ T



* T-flip flop to D-FF :

1) Excitation Table of T-FF :

Q	Q^+	T
0	0	0
0	1	1
1	0	1
1	1	0

2) characteristic Table of D-FF

D	Q	Q^+
0	0	0
0	1	0
1	0	1
1	1	1

Q	D	Q^+
0	0	0
0	1	1
1	0	0
1	1	1

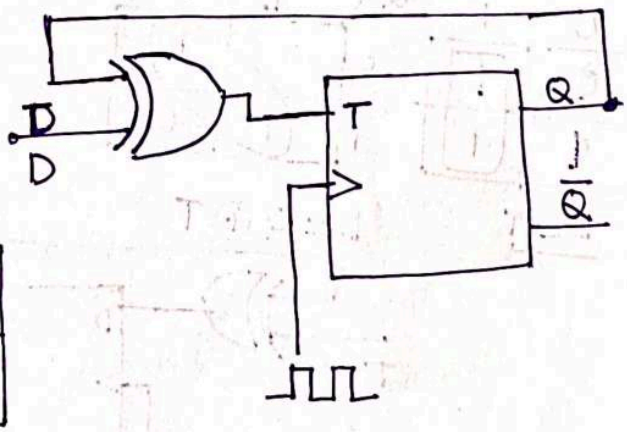
3) obtain the Boolean expression for T

Q	D	T
0	0	0
0	1	1
1	0	1
1	1	0

T

Q \ D	0	1
0	0	1
1	1	0

$T = Q \oplus D$



* JK to SR FF Conversion :- (Available → Required)

→ Excitation Table of JK FF

Q	Q^+	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

→ characteristic Table of SR FF

Q	S	R	Q^+	J	K
0	0	0	0	0	x
0	0	1	0	0	x
0	1	0	1	1	x
0	1	1	x	x	x
1	0	0	1	x	0
1	0	1	0	x	1
1	1	0	1	x	0
1	1	1	x	x	x

* Steps are required

1. Note available FF and Required FF
2. Write Excitation Table of Available FF
3. Write characteristic Table of Required FF
4. obtain bodean expression for available FF
5. Draw circuit

J

Q \ SR	00	01	11	10
0	0	0	X	1
1	X	X	X	X

S

$J = S$

K

Q \ SR	00	01	11	10
0	x	x	x	x
1	0	1	x	0

R

$K = R$

NOTE : $S = R = 1$ then $Q^+ = \overline{Q^+} = 1$ ∴ JK to SR, T to SR, D to SR not possible

* SR-FF to T-FF :-

* Steps to follow :

1) Identify available and Required FF
2) Make excitation table for available FF
3) Make characteristic table for Required FF
4) Write Boolean Expression for the available FF
5) Draw the circuit diagram.

1) Available FF → SR FF
   Required FF → T FF

2) Excitation Table of SR FF

Q	Q+	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

3) characteristic Table of T-FF

Q	T	Q+	S	R
0	0	0.	0	X
0	1	1	1	0
1	0	1	X	0
1	1	0	0	1

S

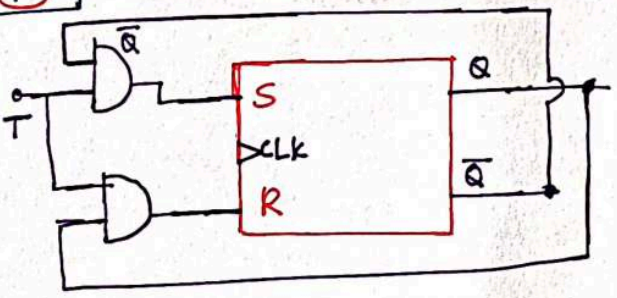
Q \ T	0	1
0	0	(1)
1	X	0

$S = \bar{Q}T$

R

Q \ T	0	1
0	X	0
1	0	(1)

$R = QT$



* SR-FF to D-FF :-

1) Available FF → SR

Required FF → D

2) Excitation Table of SR FF:

Q	Q^+	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

3) Characteristic Table of D-FF:

Q	D	Q^+	S	R
0	0	0	0	X
0	1	1	1	0
1	0	0	0	1
1	1	1	X	0

S

Q \ D	0	1
0	0	1
1	0	X

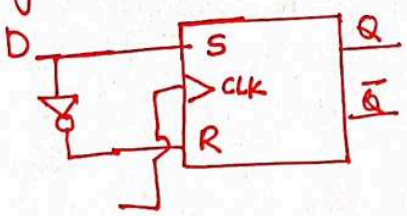
$S = D$

R

Q \ D	0	1
0	X	0
1	1	0

$R = \bar{D}$

circuit diagram



* Race around condition in JK FF:

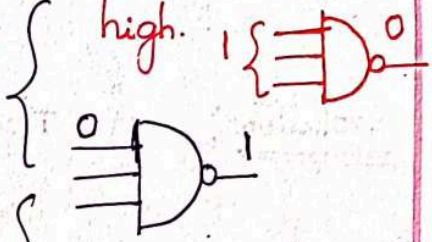
↳ when J=K=1, at that time there is possibilities of Race around condition.

↳ Truth Table of JK

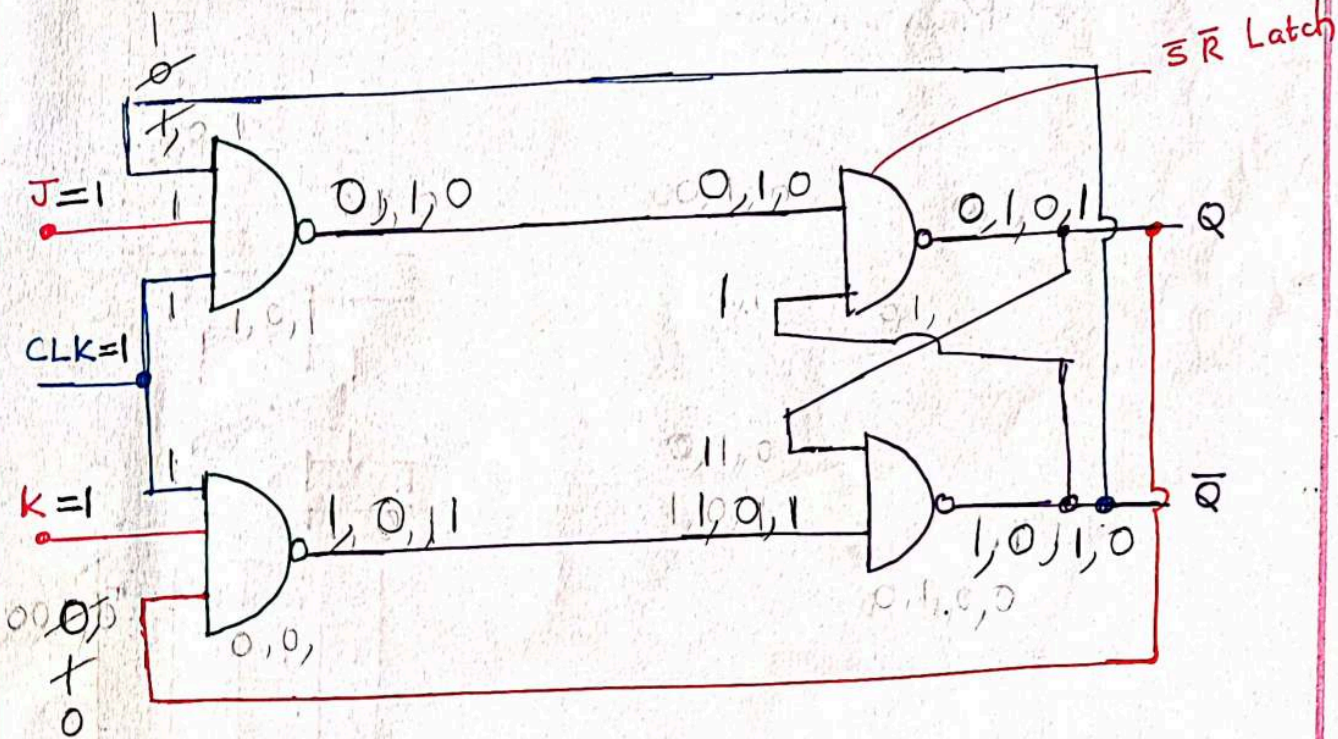
CLK	J	k	Q^+	$\bar{Q}^+$	
0	x	x	Q	$\bar{Q}$	NC Memory
1	0	0	Q	$\bar{Q}$	
1	0	1	0	1	
1	1	0	1	0	
1	1	1	$\bar{Q}$	Q	Toggling

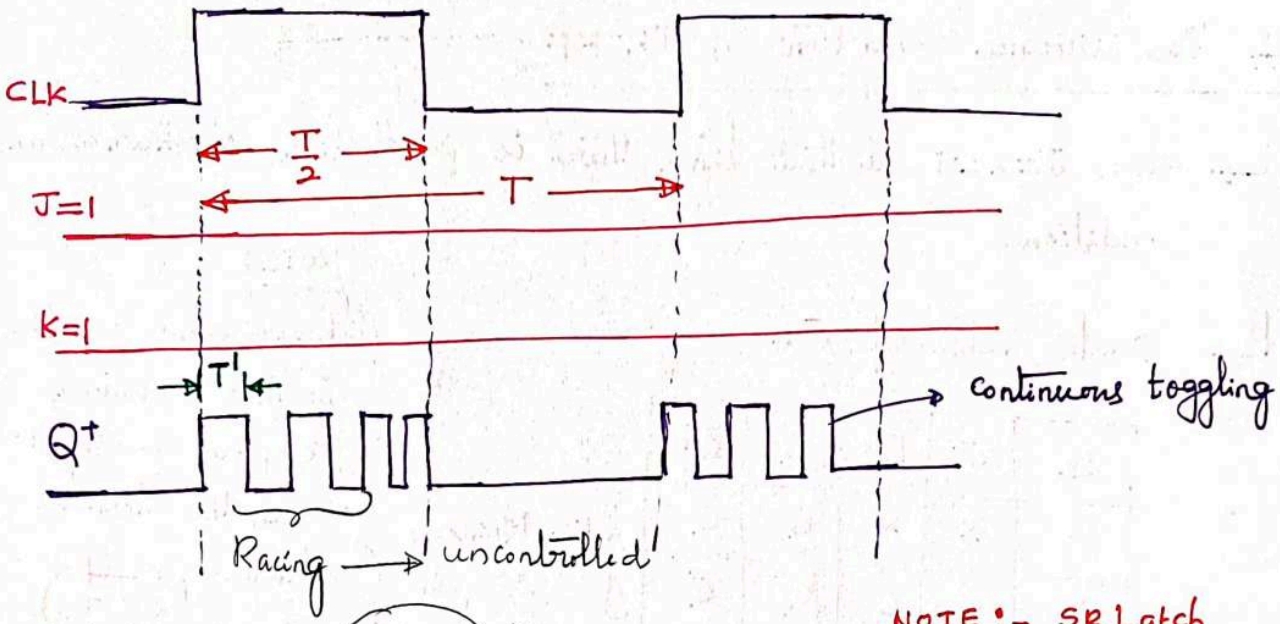
NOTE:

Race around condition is that continuous toggling of output when the clock is high.



for NAND gate, when one of the input is 0 then o/p is 1.





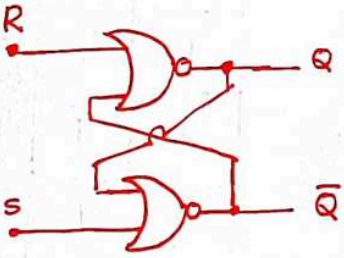
Solution:

1) $T' > \frac{T}{2}$ ✗

2) Edge Triggering Instead of pulse | Level Triggering

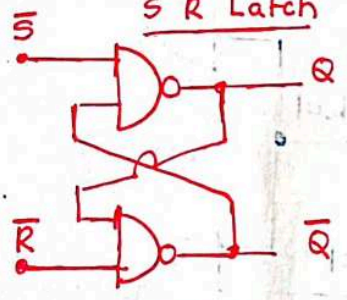
3) Master - Slave

NOTE :- SR Latch



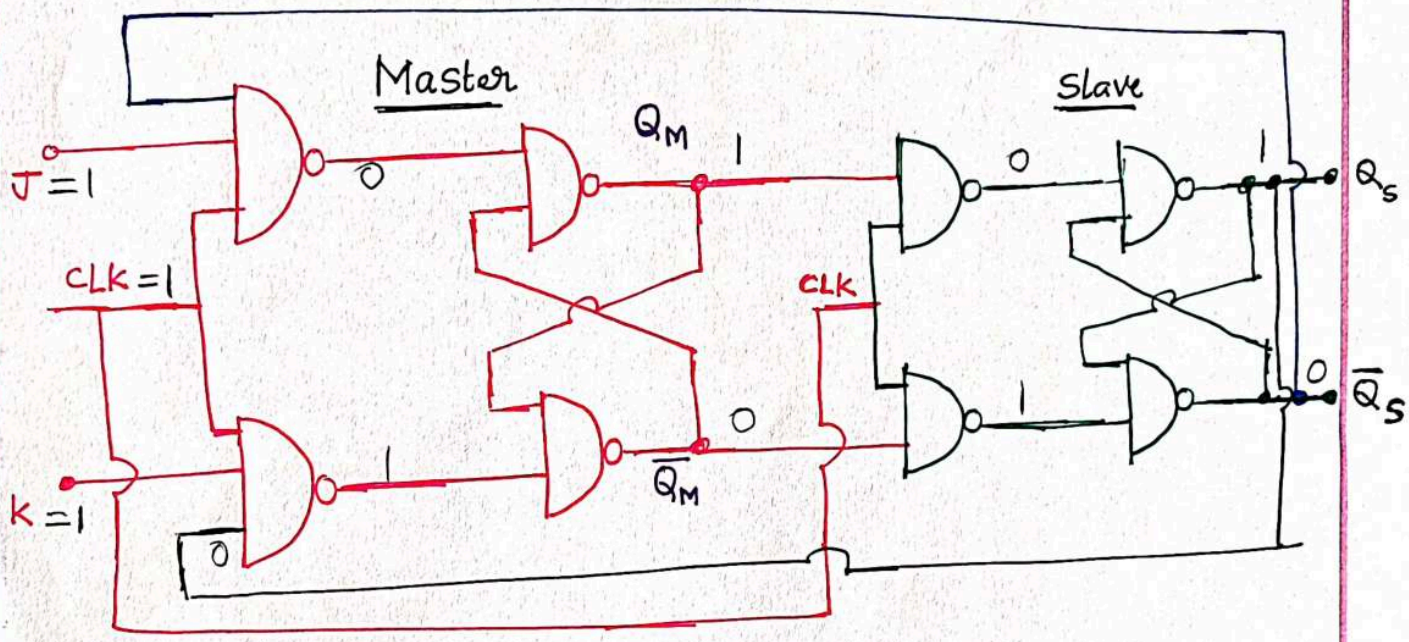
S	R	Q^+	$\bar{Q}^+$	
0	0	Q	$\bar{Q}$	
0	1	0	1	
1	0	1	0	
1	1	0*	0*	→ Invalid

$\bar{S}\bar{R}$ Latch



$\bar{S}$	$\bar{R}$	Q^+	$\bar{Q}^+$
1	1	Q	$\bar{Q}$
1	0	0	1
0	1	1	0
0	0	1*	1*

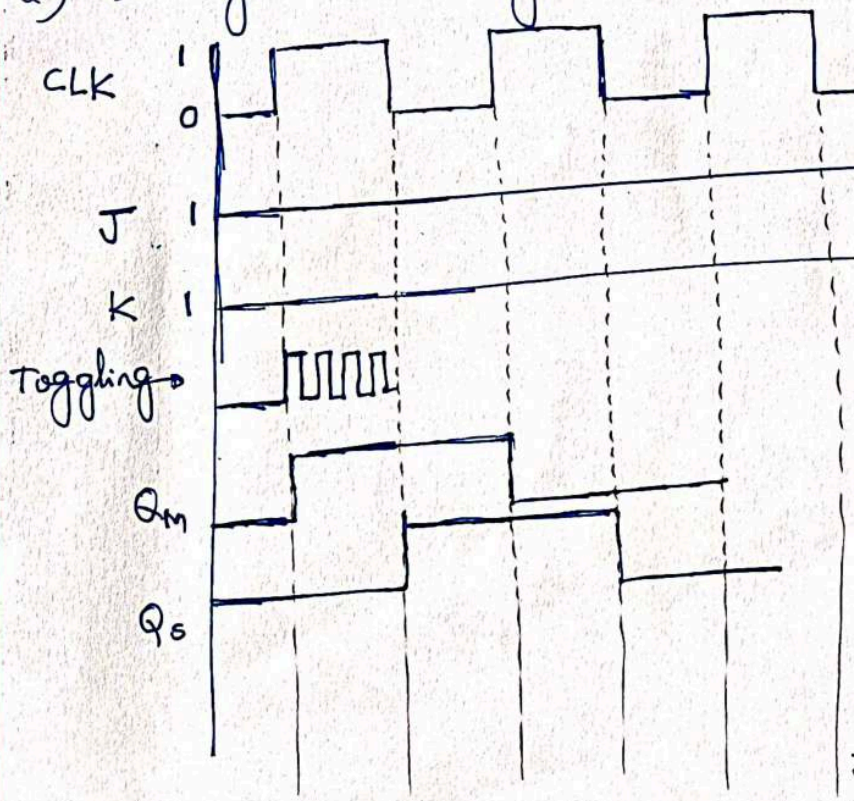
* Master - slave JK FF :—



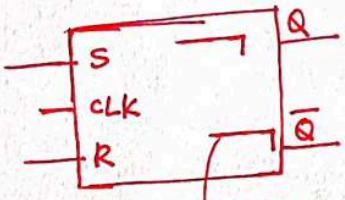
Operations

1) Assume $Q_s = 0$; $\overline{Q}_s = 1$, then $Q_M = 1$; $\overline{Q}_M = 0$ During +ve clock cycle

2) During -ve clock cycle, $Q_s = 1$ and $\overline{Q}_s = 0$



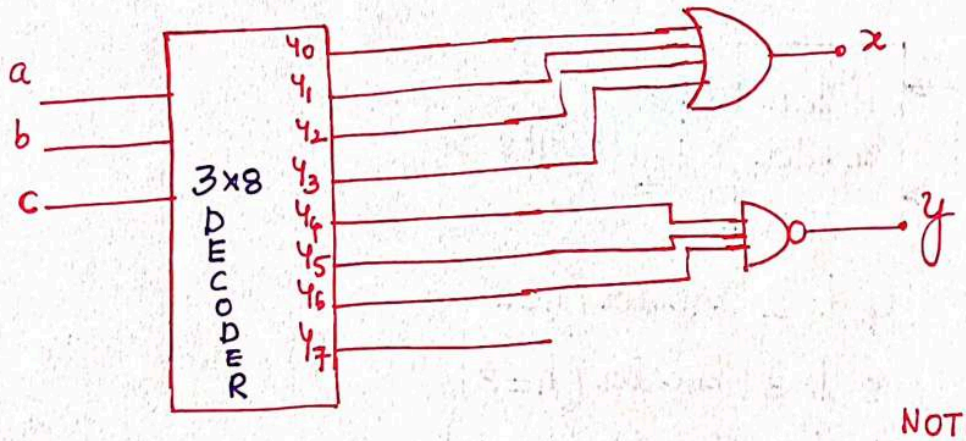
NOTE



postponed o/p indicator

The output change is postponed until the end of the pulse period

P) What is the function implemented by the given 3 to 8 Decoder?



Solution:

a	b	c	x	y
0	0	0	1	1
0	0	1	1	1
0	1	0	1	1
0	1	1	1	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	0	1

x

a \ bc	00	01	11	10
0	1	1	1	1
1	0	0	0	0

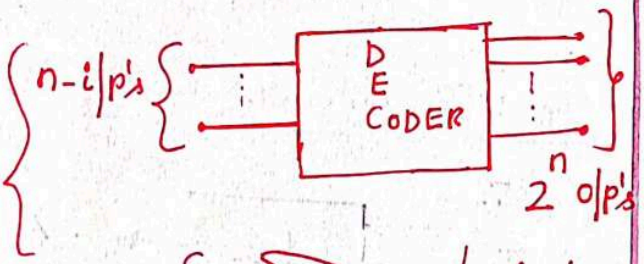
$x = \bar{a}$

y

a \ bc	00	01	11	10
0	1	1	1	1
1	1	1	1	1

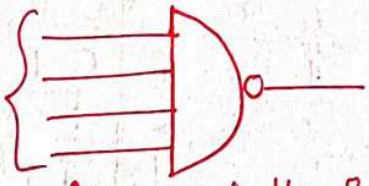
$y = 1$

NOTE





if one of the inputs is high then o/p will be high



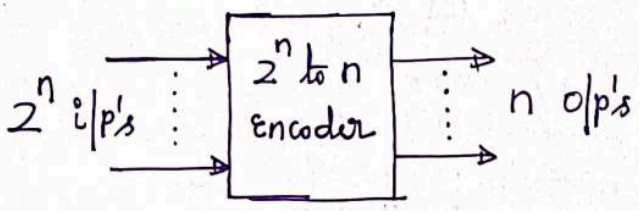
if one of the inputs is low then o/p will be high

NOTE:

$x = y_0 + y_1 + y_2 + y_3$

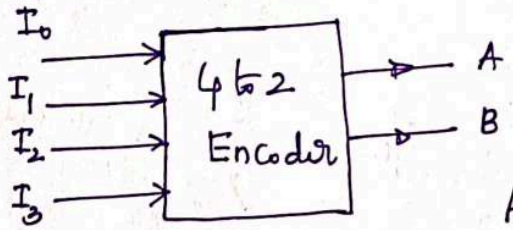
$y = \overline{y_4 \cdot y_5 \cdot y_6}$

* Encoder :



Example : 4 to 2 Encoder ($n=2$)
8 to 3 Encoder ($n=3$)
16 to 4 Encoder ($n=4$)

4 to 2 Encoder



Truth Table

I_0	I_1	I_2	I_3	A	B
1	0	0	0	0	0
0	1	0	0	0	1
0	0	1	0	1	0
0	0	0	1	1	1

A

$I_0 I_1$ \ $I_2 I_3$	00	01	11	10
00	X	1	X	1
01	0	X	X	X
11	X	X	X	X
10	0	X	X	X

I_2, I_3

$A = I_2 + I_3$

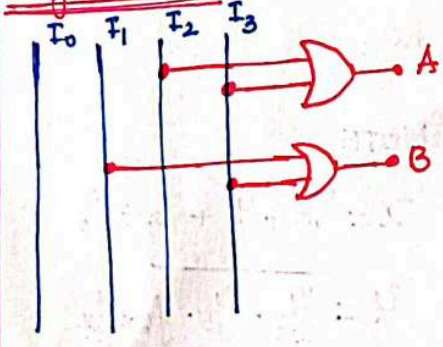
B

$I_0 I_1$ \ $I_2 I_3$	00	01	11	10
00	X	1	X	0
01	1	X	X	X
11	X	X	X	X
10	0	X	X	X

I_1, I_3

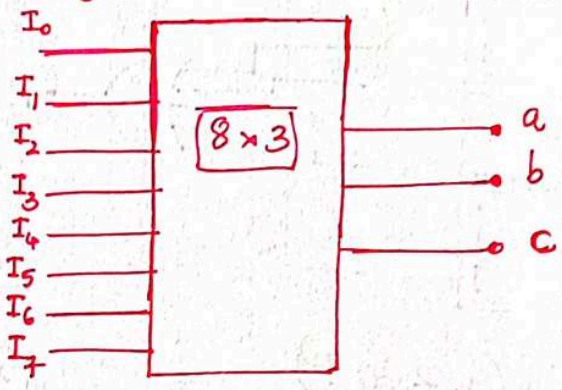
$B = I_1 + I_3$

Logic circuit :



NOTE : only one Input must be 1/ high.

~~Logic circuit~~ : Design of 8×3 encoder



* Truth Table of 8×3 encoder :

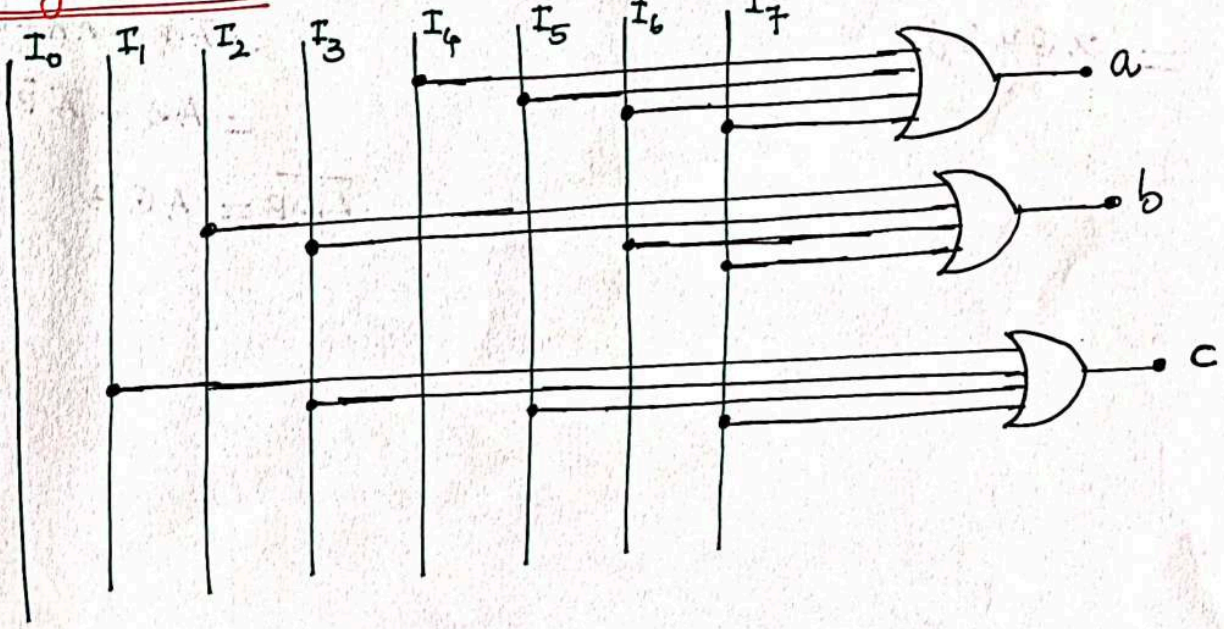
I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7	a	b	c
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
1	0	0	0	0	0	0	1	1	1	1

$a = I_4 + I_5 + I_6 + I_7$

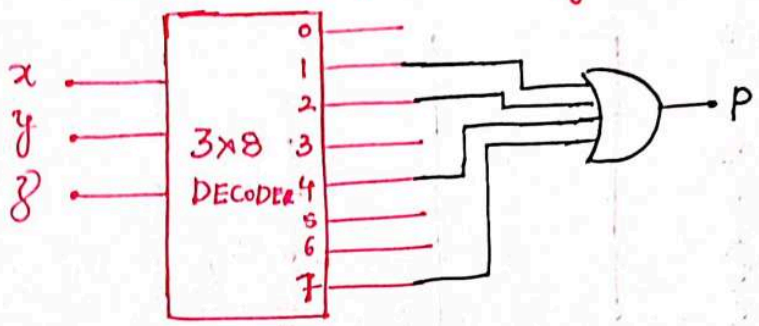
$b = I_2 + I_3 + I_6 + I_7$

$c = I_1 + I_3 + I_5 + I_7$

* Logic circuit



P) What is the function Implemented by the below circuit



Solution : $P = \sum m(1, 2, 4, 7)$

P	yz 00	01	11	10
x = 0	0	(1)	0	(1)
x = 1	(1)	0	(1)	0

x y z	P
0 0 0	0
0 0 1	1
0 1 0	1
0 1 1	0
1 0 0	1
1 0 1	0
1 1 0	0
1 1 1	1

$\overline{x+y} = \bar{x} \cdot \bar{y}$

$\overline{xy} = \bar{x} + \bar{y}$

$P = \bar{x}\,\bar{y}\,\bar{z} + \bar{x}\,y\,\bar{z} + x\,\bar{y}\,\bar{z} + x\,y\,z$

$= \bar{z}(xy + \bar{x}\,\bar{y}) + \bar{z}(\bar{x}\,y + x\,\bar{y})$

$= \bar{z}(x \odot y) + \bar{z}(x \oplus y)$

$= z\,\overline{(x \oplus y)} + \bar{z}(x \oplus y)$

(a, b, a, b)

$= x \oplus y \oplus z$

$A \oplus B = A\bar{B} + B\bar{A}$

$\overline{A \oplus B} = \overline{A\bar{B} + B\bar{A}}$

$= \overline{A\bar{B}} \cdot \overline{B\bar{A}}$

$= (\bar{A} + \bar{\bar{B}})(\bar{B} + \bar{\bar{A}})$

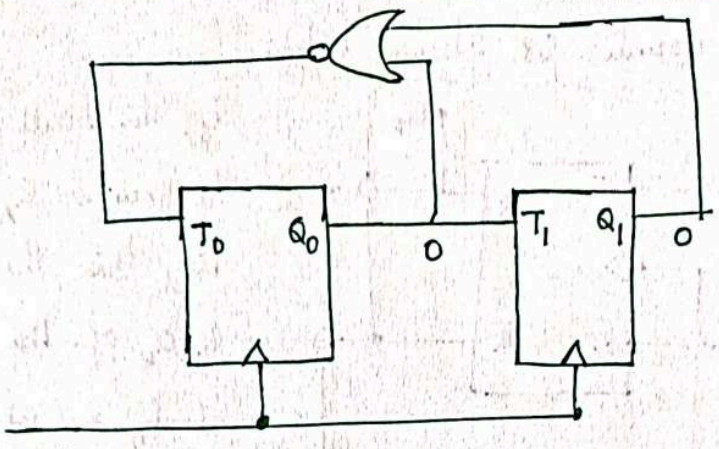
$= (\bar{A} + B)(\bar{B} + A)$

$= \bar{A}\bar{B} + \bar{A}A + B\bar{B} + AB$

$= AB + \bar{A}\bar{B}$

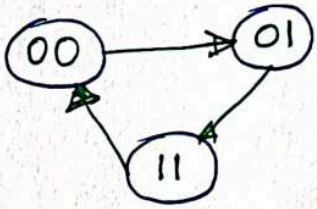
$\overline{A \oplus B} = A \odot B$

Q)

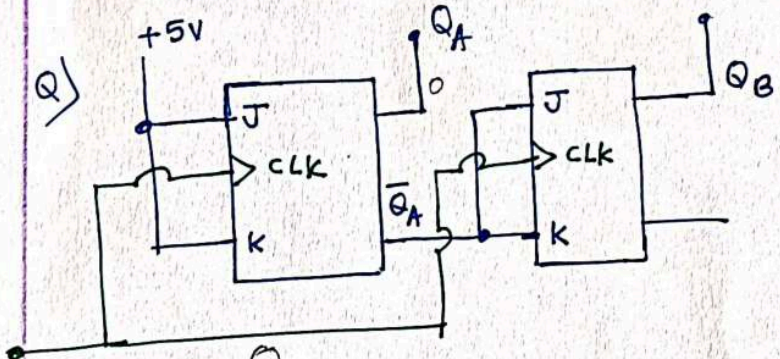


Present state		Next state	
Q_1	Q_0	Q_1^+	Q_0^+
0	0	0	1
0	1	1	1
1	0	0	0
1	1	0	0

Q_1	Q_0
0	0
0	1
1	1
0	0



Q)

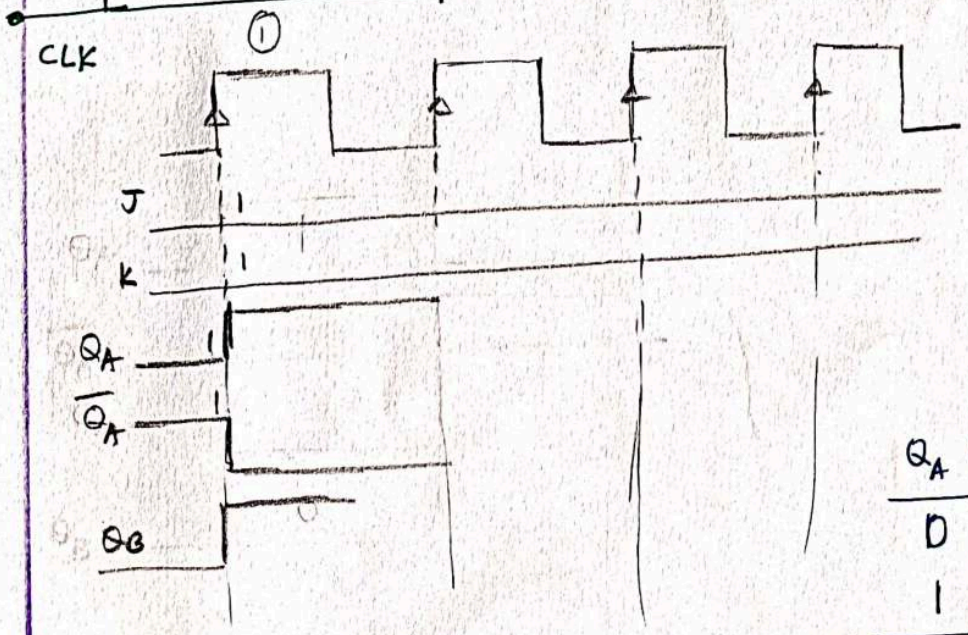


The current state of $Q_A Q_B$ is 00.

The next state will be

NOTE

$Q^+ = J\bar{Q} + \bar{K}Q$



$$Q_A^+ = \bar{Q}_A$$

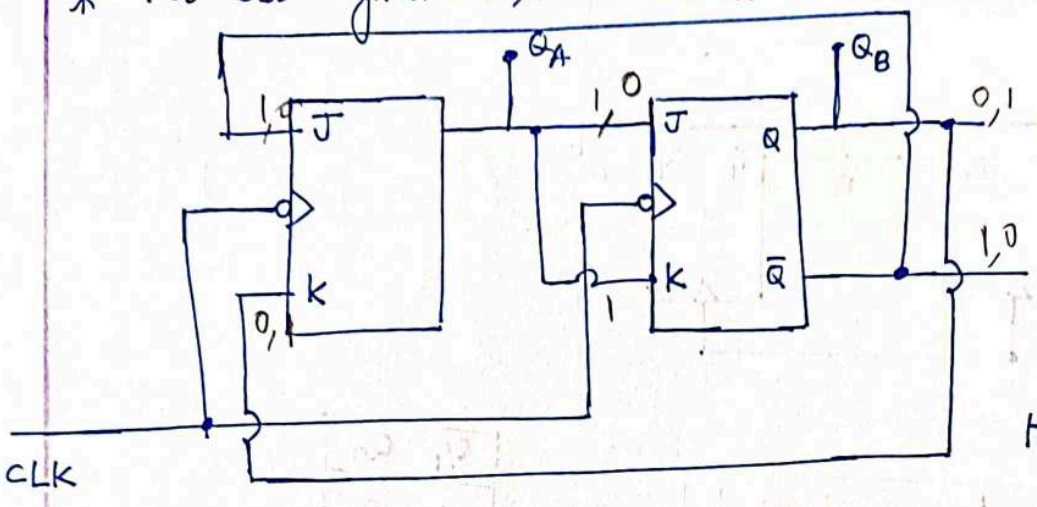
$$Q_B^+ = J_B\bar{Q}_B + \bar{K}_B Q_B$$

$$= \bar{Q}_A\bar{Q}_B + \bar{Q}_A Q_B$$

$$Q_B^+ = Q_A \odot Q_B$$

Q_A	Q_B
0	0
1	1

* Two bit synchronous counter is shown below



if the state of the counter $Q_A Q_B$ is 10, then the state $Q_A Q_B$ of the counter at t_{n+3} (after three clock pulses) will be

Q_A	Q_B
1	0
↑ 0	1

$Q^+ = J\bar{Q} + Q\bar{K}$ $\quad X = 0,\ \bar{X} = 1$

$Q_A = J_A \bar{Q}_A + Q_A \bar{K}_A$ $\quad X + \bar{X} = 1$

$= \bar{Q}_B \bar{Q}_A + Q_A \bar{Q}_B$ $\quad X \cdot \bar{X} = 0$

$= \bar{Q}_B (\bar{Q}_A + Q_A)$

$\boxed{Q_A = \bar{Q}_B}$

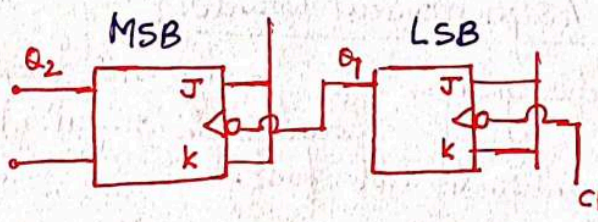
Q_A	Q_B
1	0
↑ 1	1
↑ 0	0
↑ 1	0

$Q_B = J_B \bar{Q}_B + Q_B \bar{J}_B$

$= Q_A \bar{Q}_B + Q_B \bar{Q}_A$

$= Q_A \oplus Q_B$

Q)

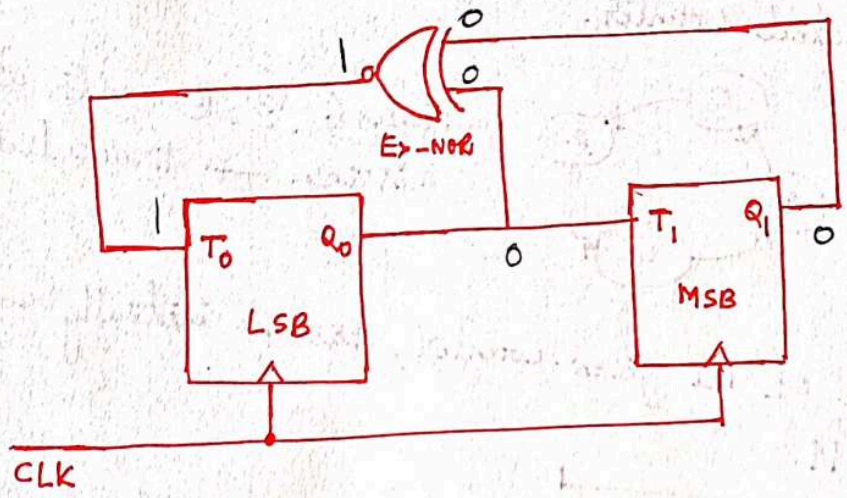


What type of counter?

Ans: Asynchronous up counter

NOTE: In Asynchronous counter, the FF which is connected to external clock will be considered as LSB.

Q)

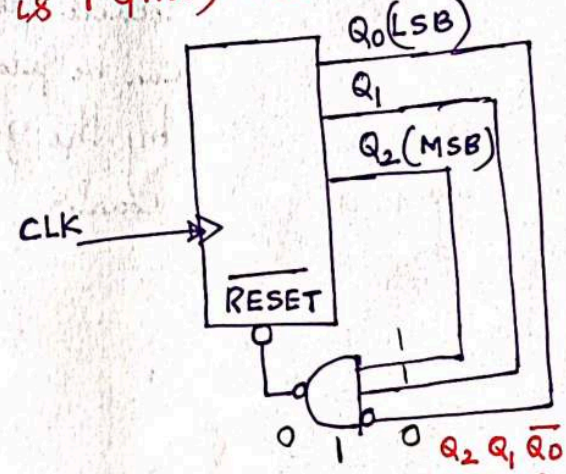


NOTE			
	Ex-OR	Ex-NOR	
A	B	A⊕B	A⊙B
0	0	0	1
0	1	1	0
1	0	1	0
1	1	0	1

Q_1	Q_0	Ex-NOR
0	0	1
0	1	0
1	1	1
1	0	0
0	0	1

A counter is said to be self-starting, if it is possible to enter counting loops irrespective of the initial state. A counter is said to be free running if it contains all possible states in the counting loop. So the given counter is neither self-starting nor free running.

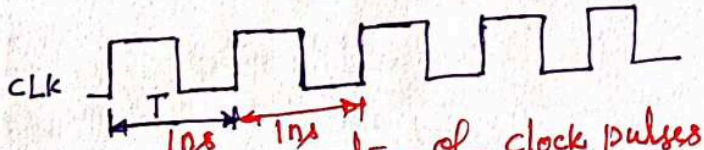
Q) For the below circuit, the delay of the bubbled NAND gate is 2ns and that of the counter is assumed to be zero. If the clock frequency is 1 GHz, then the counter behaves as a _____



Given, $f_{CLK} = 1\,GHz$

$T_{CLK} = \frac{1}{f_{CLK}} = \frac{1}{10^9} = 10^{-9}\,s$

$= 1\,ns$



Counter → Counting of clock pulses

If no delay in the gate then it's a mod-6 counter, because of delay of gate is 2ns then it becomes MOD-8 counter

Q) What are the demerits of the Asynchronous Counter?

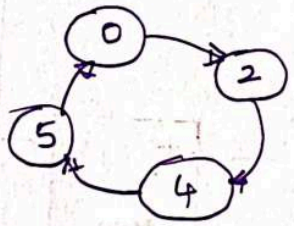
↳ Random Counter cannot be designed (only for up and down counter in sequence)

↳ speed of Asynchronous counter is not fast.

Q) A counter that counts sequentially but does not step through all possible states, it returns to zero after a particular state

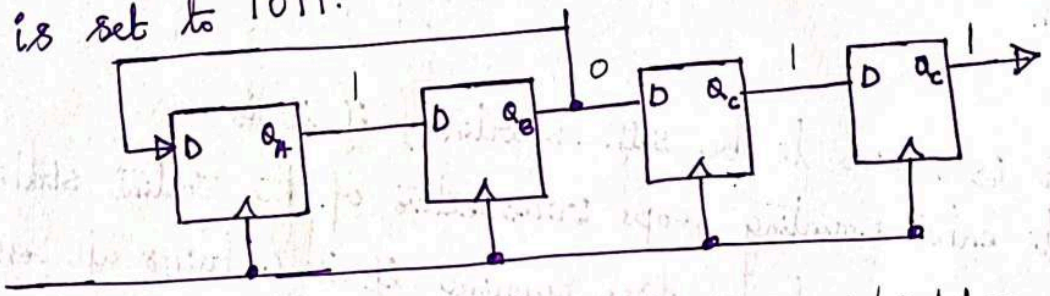
Ans: Truncated Counter.

Ex:



states ①, ③, ⑥ and ⑦ are missing → Truncated Counter

Q) The mod number of the given counter, Assume initially counter is set to 1011.



1011
0101
1010
0100
1010
0101
1000
0100
1010
0101

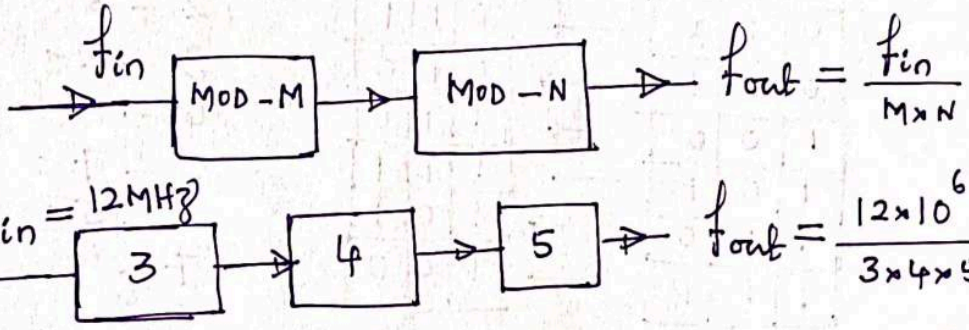
1011
0101
1010
0101
1011

0 → 1011
↑ — 0101
↑ — 1010
↑ — 0101
↑ — 1010
↑ — 0101
↑ — 1010
↑ —

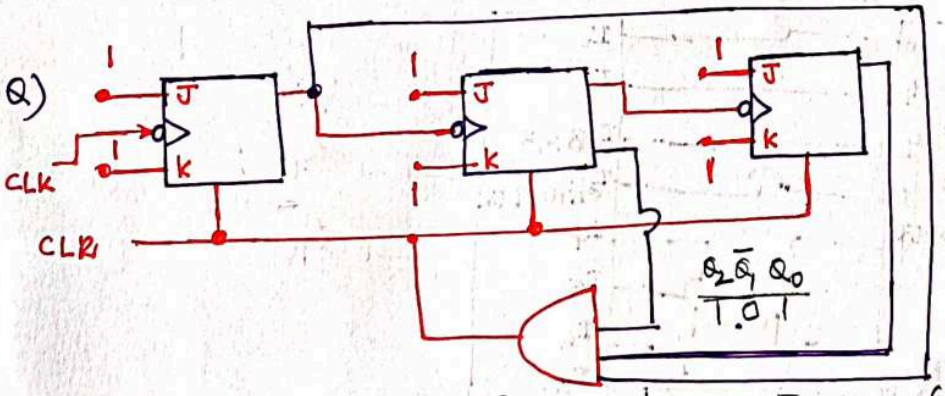
Ans: only two locked states can be produced by the given counter

Q) 12 MHz clock frequency is applied to a cascaded counter of modulus-3 (MOD-3) counter, modulus-4 counter and modulus-5 counter, what are the lowest o/p frequency and overall modulus, respectively?

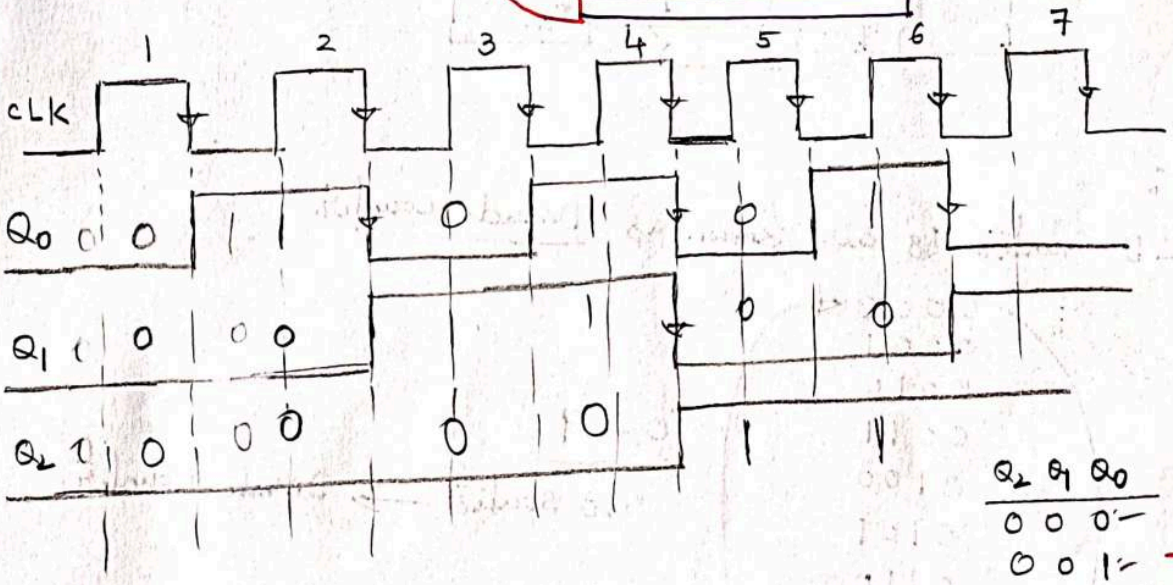
Solution:



Q)



which type of Counter?

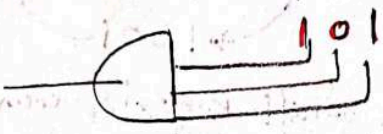


NOTE:

ϕ → up count

⊦ → Down

Q2 Q1 Q0	
0 0 0	
0 0 1	– → up
0 1 0	+ → Down
0 1 1	
1 0 0	
1 0 1	

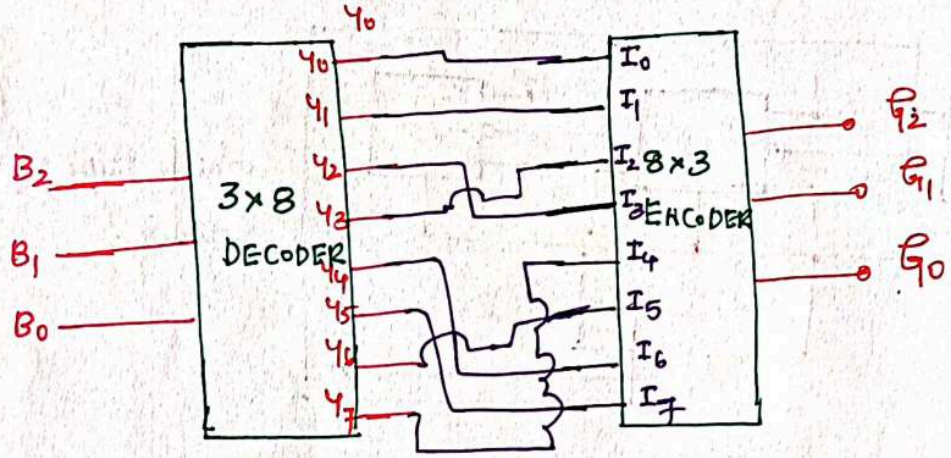


Ans: MOD-5, up counter

* Binary to Gray code converter using Decoder and Encoder :

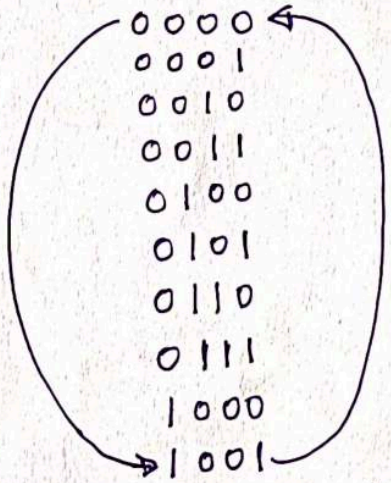
B_2	B_1	B_0	G_2	G_1	G_0
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	1
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	1	1	1
1	1	0	1	0	1
1	1	1	1	0	0

0 — 0
1 — 1
2 — 3
3 — 2
4 — 6
5 — 7
6 — 5
7 — 4



NOTE:

1) BCD counter is also known as Decade counter



0 – 9
10 states → Decade Counter

2) Which shift Register counter Requires the most-decoding circuitry?

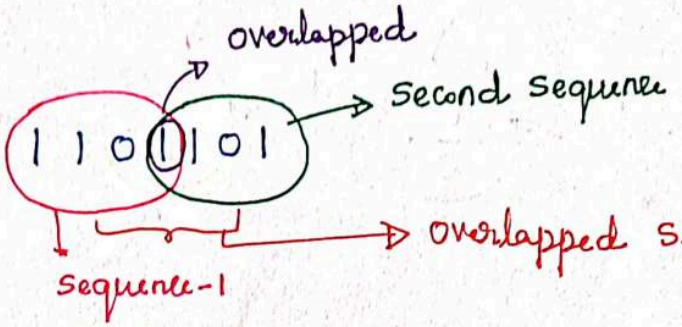
↳ Asynchronous Counter / Ripple Counter

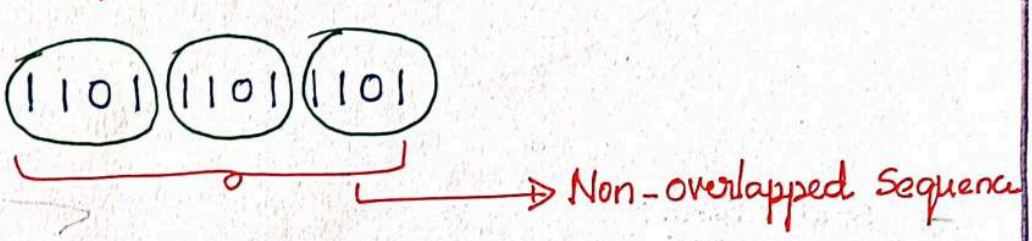
Q) Design a state diagram of the 1101 sequence if it is output goes to 1 when a target sequence has been detected.

Solution:

NOTE:
overlapping | Non-overlapping.

Example:



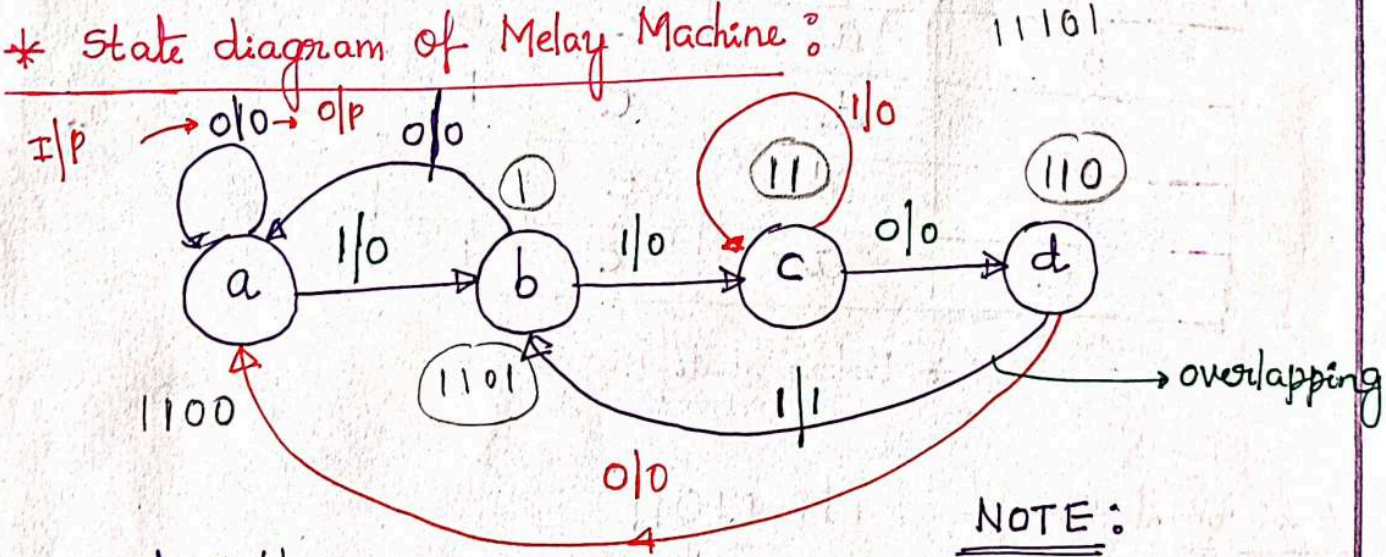


1) overlapping:

Target sequence: 1101

NOTE:
0 → No Transition
1 → Transition

* State diagram of Melay Machine:



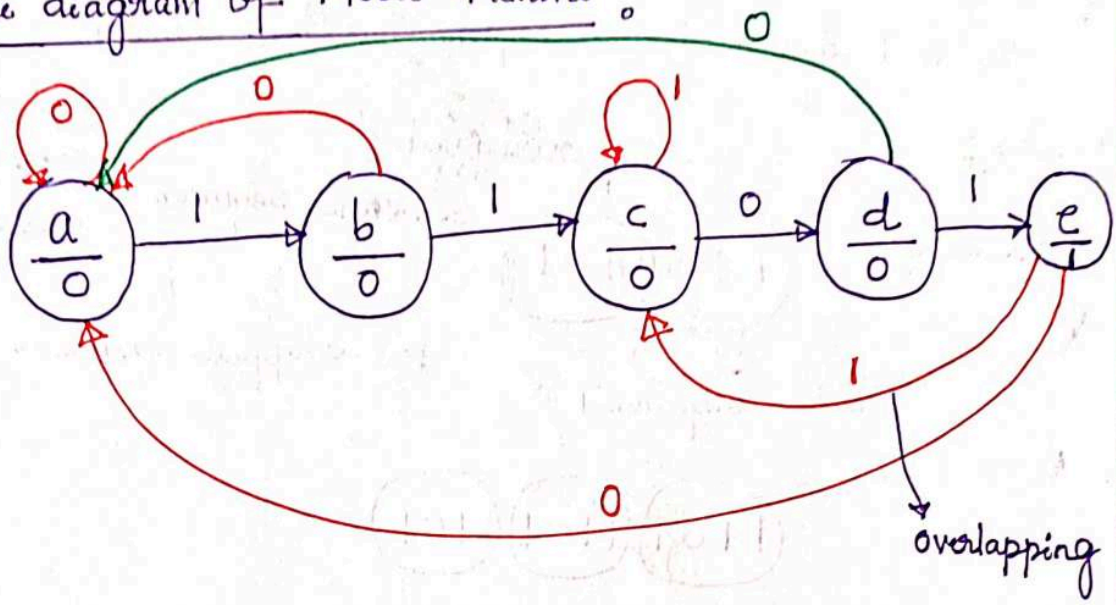
NOTE:

I/P seq = 1 1 0 1 1 0 1
O/P = 0 0 0 1 0 0 1

* State Table

present state	Next state		output	
	i/p=0	i/p=1	i/p=0	i/p=1
a	a	b	0	0
b	a	c	0	0
c	d	c	0	0
d	a	b	0	1

* State diagram of Moore Machine :



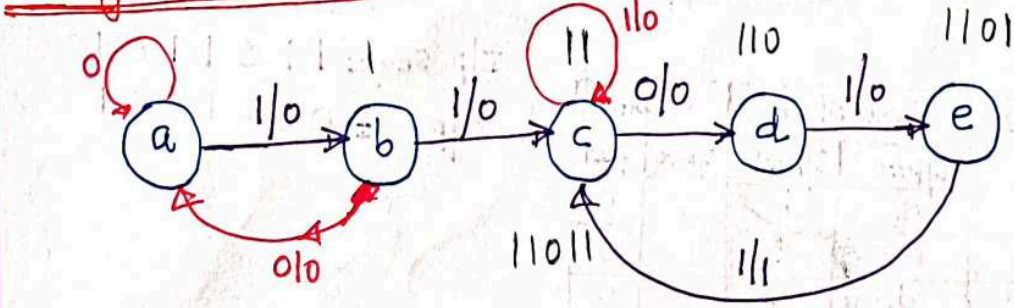
* State Table :

present state	Next state		O/p
	Input = 0	I/P = 1	
a	a	b	0
b	a	c	0
c	d	e	0
d	a	e	0
e	a	c	1

→ Extra state / hardware

1101101

Assignment Question : 1101101

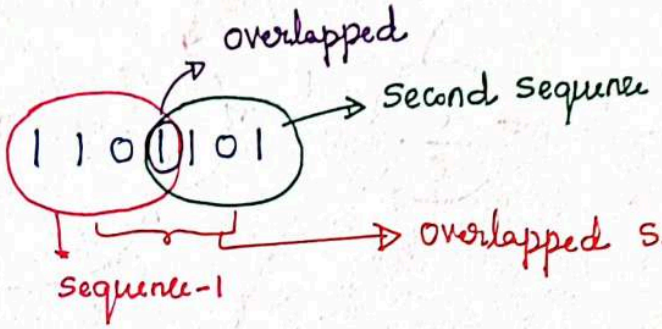


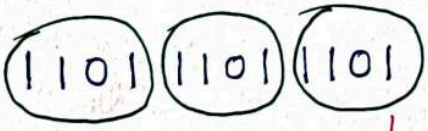
Q) Design a state diagram of the 1101 sequence if it is output goes to 1 when a target sequence has been detected.

Solution:

NOTE:
overlapping | Non-overlapping

Example:





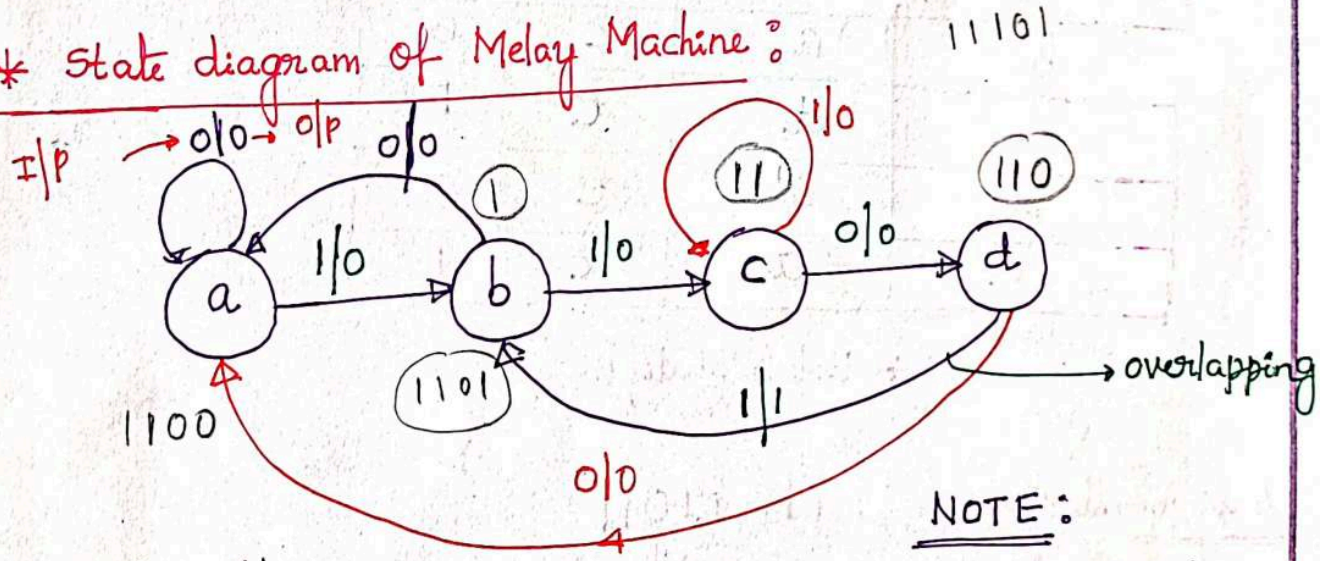
Non-overlapped Sequence

1) overlapping:

Target sequence: 1101

NOTE:
0 → No Transition
1 → Transition

* State diagram of Melay Machine:



* State Table

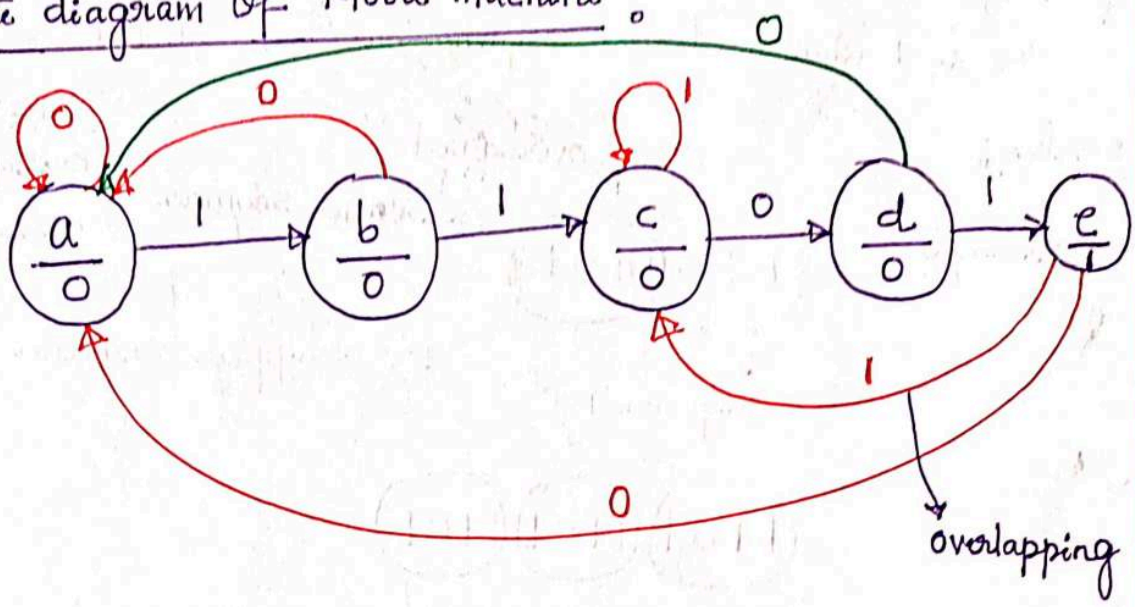
present state	Next state		output	
	i/p=0	i/p=1	i/p=0	i/p=1
a	a	b	0	0
b	a	c	0	0
c	d	c	0	0
d	a	b	0	1

NOTE:

I/p seq = 1 1 0 1 1 0 1

O/p = 0 0 0 1 0 0 1

* State diagram of Moore Machine :



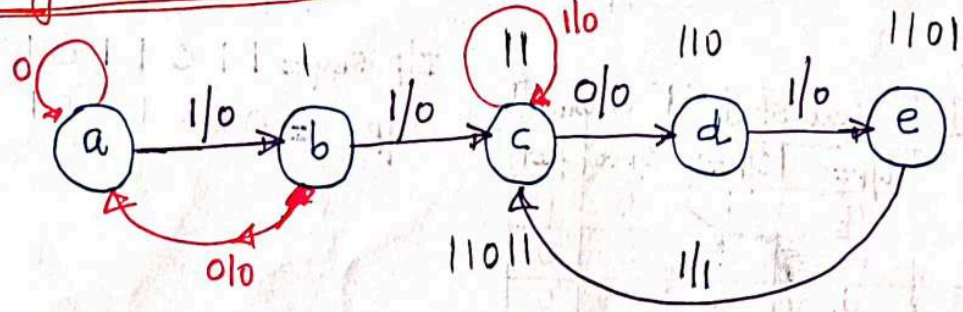
* State Table :

present state	Next state		o/p
	Input = 0	I/p = 1	
a	a	b	0
b	a	c	0
c	d	e	0
d	a	e	0
e	a	c	1

→ Extra state / hardware

1101101

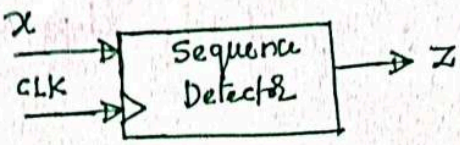
Assignment Question : 1101101



* Mealy type sequence detector:

→ Design a Mealy circuit to detect a sequence if the input $x = 111$ then output $Z = 1$ otherwise $Z = 0$.

Solution:



Example: overlapping

$x = i/p = 1\,0\,1\,1\,1\,0\,0\,1\,0\,1\,1\,1\,1\,0$

$Z = o/p = 0\,0\,0\,0\,1\,0\,0\,0\,0\,0\,0\,0\,1\,0$

NOTE:

Moore circuit

$n \longrightarrow n+1$

no. of bits — no. of states

Ex: 3 → 4

Mealy circuit

$n \longrightarrow n$

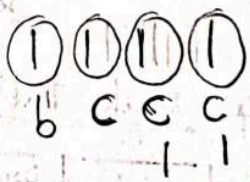
no. of bits — no. of states

Ex: 3 → 3

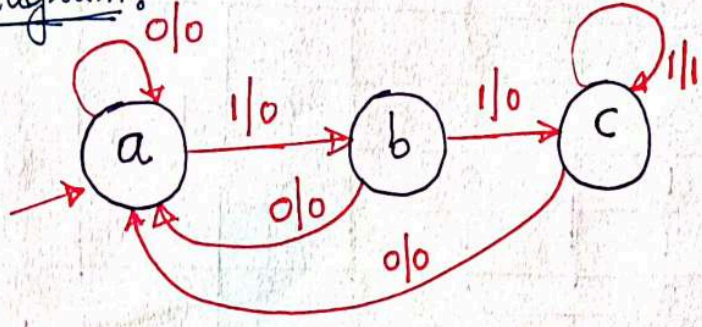
* Mealy machines typically use less states than Moore machines

→ Steps for a sequence detector

1) State diagram
2) Find number of flip flops
3) State Assignment
4) State Table
5) K-map Simplification
6) circuit Design.



1) State diagram:



2) To find no. of FF's

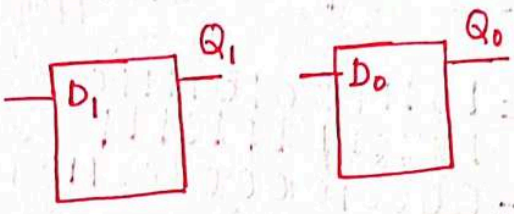
We have, $2^n \geq$ No. of states (n → no. of flip-flops)

$2^2 \geq 3$

$4 \geq 3$

∴ No. of flip-flops required = 2

Example:



NOTE:

{ present state = $Q_1 Q_0$
{ Next state = $D_1 D_0$

{ $Q_1 Q_0$ → D FF o/p's
{ $D_1 D_0$ → D FF i/p's

3. State Assignment Table:

Since two FFs are required, ∴ Two bits are required to represent the state.

2 FF's → 2 bits
3 FF's → 3 bits
4 FF's → 4 bits

State Assignment

a = 00
b = 01
c = 10
(11) unused

4. State Table:

state	present State		present i/p	Next state		o/p
	Q_1	Q_0	x	Q_1^+	Q_0^+	Z
a	0	0	0	0	0	0
	0	0	1	0	1	0
b	0	1	0	0	0	0
	0	1	1	1	0	0
c	1	0	0	0	0	0
	1	0	1	1	0	1

5) K-Map Simplification:

$Q_1^+ = D_1$

x \ Q_1Q_0	00	01	11	10
0	0	0	X	0
1	0	1	X	1

xQ_0 xQ_1

$$Q_1^+ = xQ_0 + xQ_1 = D_1$$

$Q_0^+ = D_2$

x \ Q_1Q_0	00	01	11	10
0	0	0	X	0
1	1	0	X	0

$x\bar{Q}_1\bar{Q}_0$

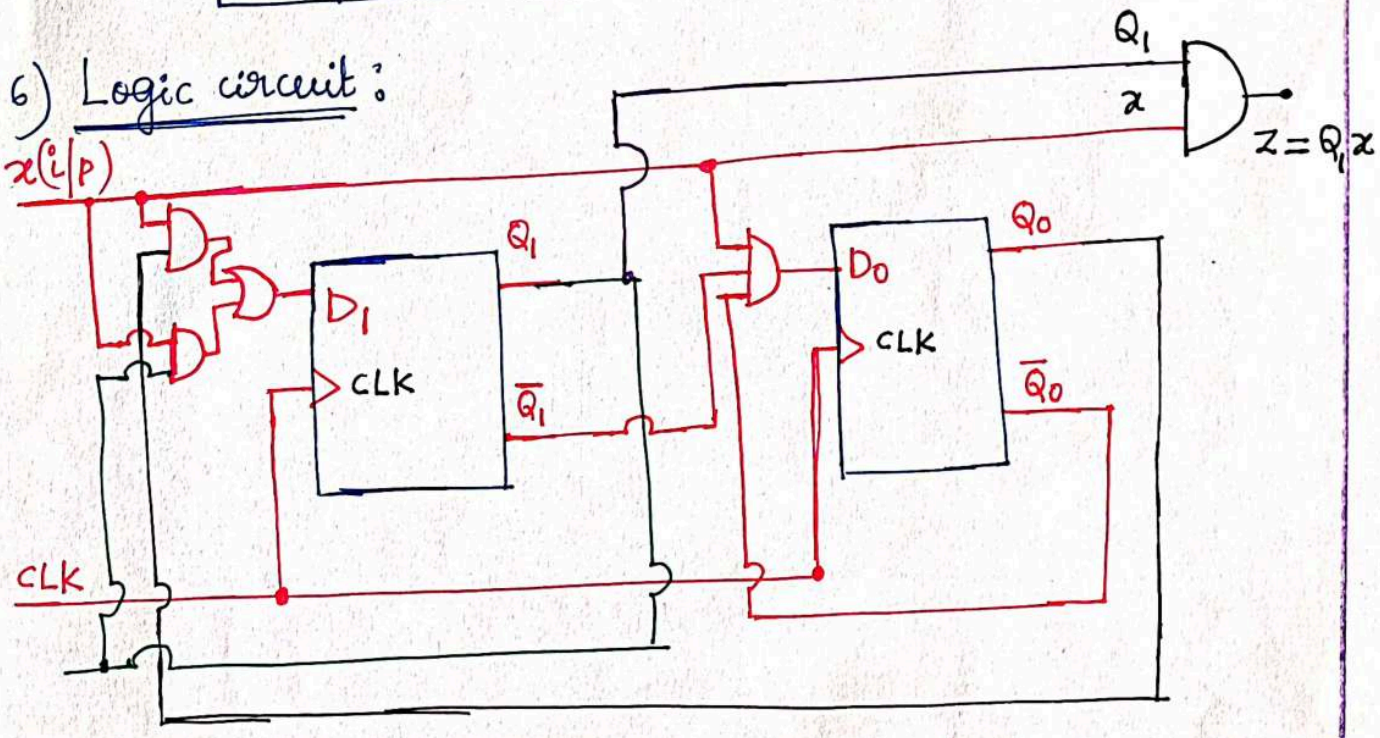
$$Q_0^+ = x\bar{Q}_1\bar{Q}_0 = D_0$$

Z

x \ Q_1Q_0	00	01	11	10
0	0	0	X	0
1	0	0	X	1

$$Z = xQ_1$$

6) Logic circuit:

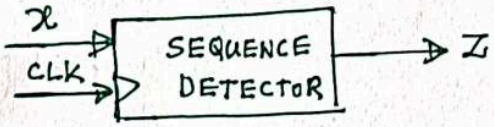


NOTE: Synchronous clock

* Moore Sequence detector :-

→ Design a Moore circuit to detect a sequence if the input $x=111$ then output $Z=1$ otherwise $Z=0$.

Solution :

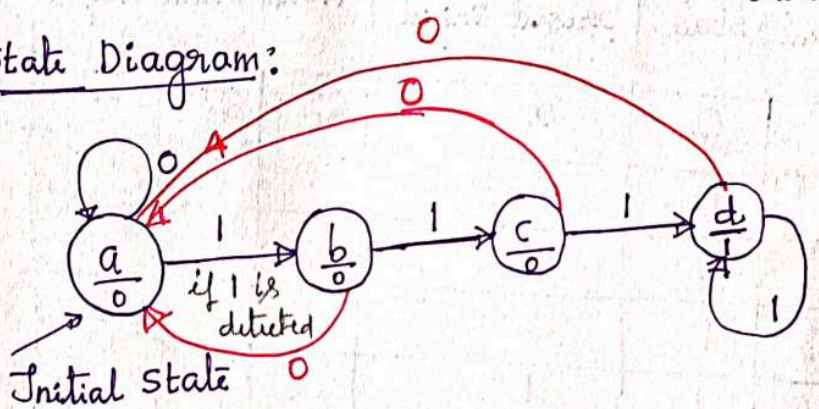


Ex: i/p ; x = 1 0 1 1 1 0 0 1 0 1 1 1 1 0

o/p ; Z = — — 1 — — — 1 1 —

→ A sequence detector is a sequential circuit detects a specific sequence of bits in the input

1. State Diagram:



NOTE:

Initial state → a: 0 → a, 1 → b

b: 1 → 0 → a, 1 → c

c: 11 → 0 → a, 1 → d

d: 111 → 0 → a, 1 → d

NOTE:

Moore circuit

n → n+1

n : no. of bits ; n+1 : no. of states

Ex: 3 → 4

2. To find no. of FF's:

$2^n \geq$ No. of states → no. of flip-flops | No. of bits

We have, $2^n \geq$ No. of States

$2^2 \geq 4$

$\boxed{4 \geq 4}$

∴ No. of flip-flops required = 2 ;

3. State Assignment Table:

Since two FFs are required, ∴ Two bits are required to represent the state.

State Assignment:

a = 00
b = 01
c = 10
d = 11

NOTE:

$n = \log_2 m$

m → no. of states

n → Minimum of state bits.

4. State Table

State	present state		present Input	Next state		output
	Q_1	Q_0	x	Q_1^+	Q_0^+	Z
a	0	0	0	0	0	0
	0	0	1	0	1	0
b	0	1	0	0	0	0
	0	1	1	1	0	0
c	1	0	0	0	0	0
	1	0	1	1	1	0
d	1	1	0	0	0	1
	1	1	1	1	1	1

Z

5. K-Map Simplification

$Q_1^+ = D_1$

x \ Q_1Q_0	00	01	11	10
0	0	0	0	0
1	0	1	1	1

xQ_0 xQ_1

$Q_1^+ = D_1 = xQ_0 + xQ_1$

$Q_0^+ = D_0$

x \ Q_1Q_0	00	01	11	10
0	0	0	0	0
1	1	0	1	1

$Q_0^+ = x\bar{Q}_0 + xQ_1$

Z

x \ Q_1Q_0	00	01	11	10
0	0	0	1	0
1	0	0	1	0

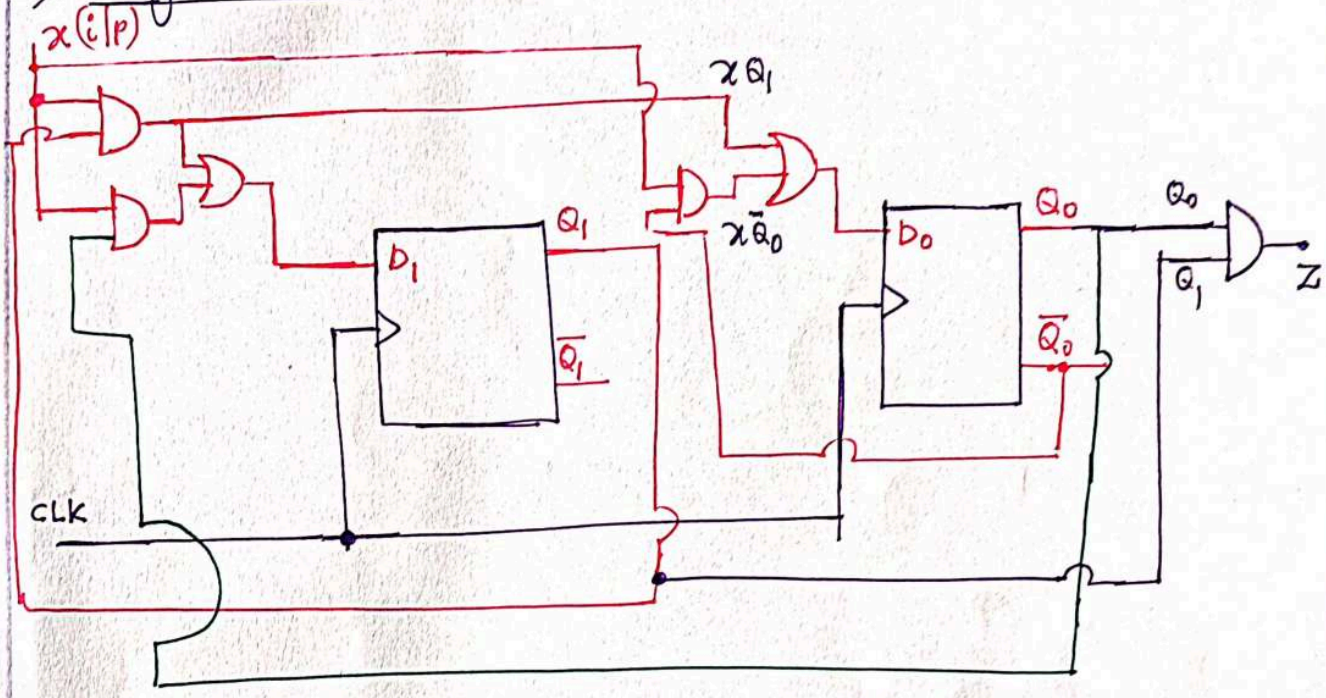
$Z = Q_1Q_0$

(or)

Q_1 \ Q_0	0	1
0	0	0
1	0	1

$Z = Q_1 Q_0$.

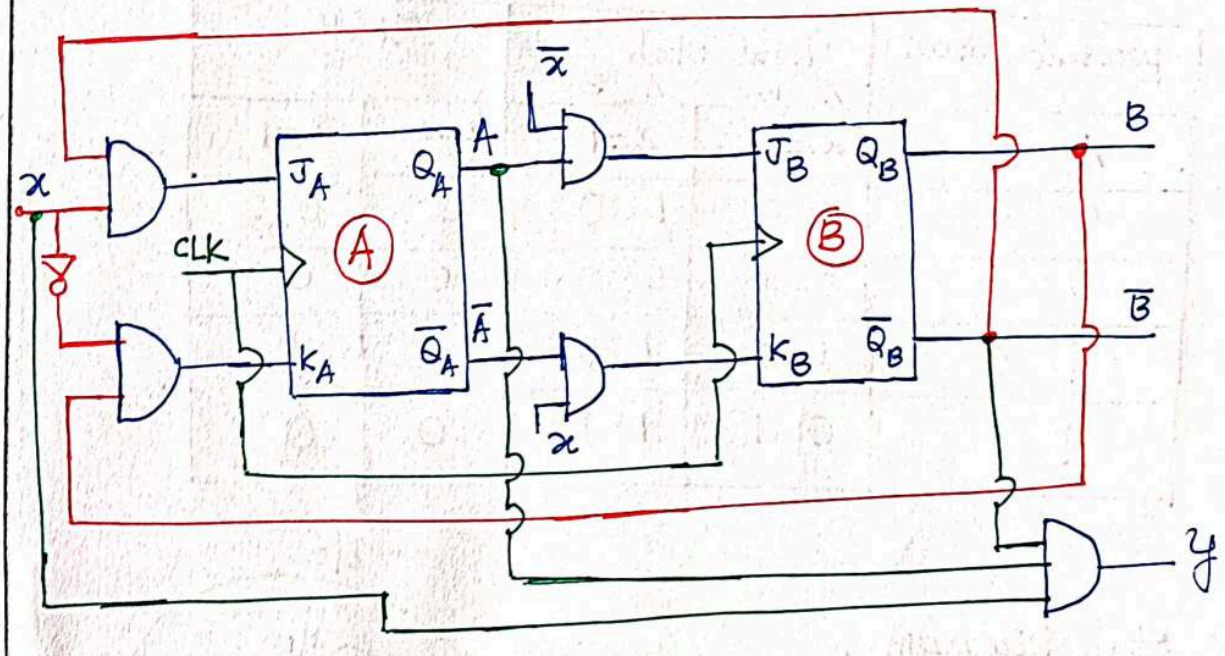
6) Logic circuit:



NOTE: Synchronous clock

* Analysis of synchronous sequential circuit

Given circuit:



Step1: Determine the equations for the flip-flop inputs and the equation for the output (y).

$J_A = x\bar{B}$ $\quad J_B = \bar{x}A$

$K_A = \bar{x}B$ $\quad K_B = x\bar{A}$

NOTE

$\overline{AB} = \bar{A} + \bar{B}$

We have characteristic equation of JK FF

$$Q^+ = \bar{Q}J + Q\bar{K}$$

Apply to 'A' FF

$$Q_A^+ = J_A\bar{Q}_A + \bar{K}_A Q_A$$

$$= x\bar{B}\bar{A} + \overline{\bar{x}B}\,A$$

$$= x\bar{A}\bar{B} + (\bar{\bar{x}} + \bar{B})\cdot A$$

$$= x\bar{A}\bar{B} + xA + A\bar{B}$$

Next state Equation of A FF

$$\boxed{Q_A^+ = x\bar{A}\bar{B} + Ax + A\bar{B}}$$

Apply to B FF → Next state Equation

$$Q_B^+ = J_B\bar{Q}_B + \bar{K}_B Q_B$$

$$= \bar{x}A\bar{B} + \overline{x\bar{A}}\,B$$

$$= A\bar{B}\bar{x} + (\bar{x} + \bar{\bar{A}})B$$

$$Q_B^+ = A\bar{B}\bar{x} + B\bar{x} + AB$$

$$\boxed{Q_B^+ = A\bar{B}\bar{x} + AB + B\bar{x}}$$

Next state Equation of 'B' FF

$$Y = x A \bar{B}$$

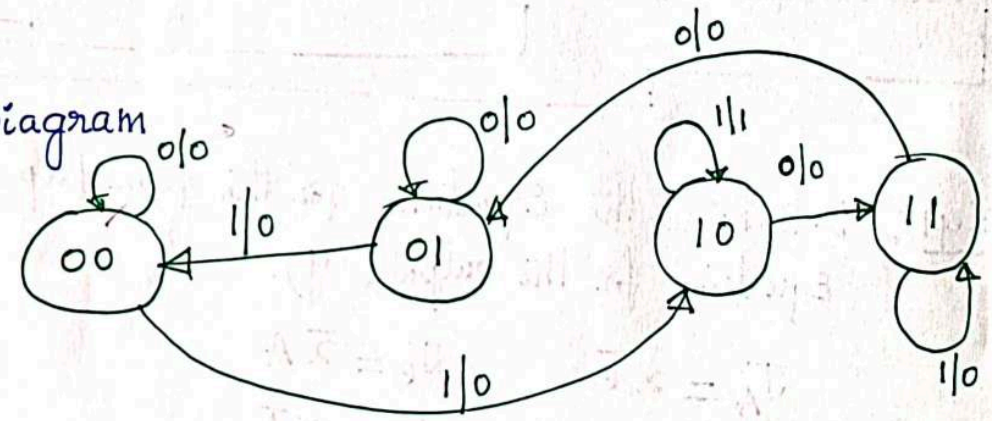
$$Q_A^+ = x\bar{A}\bar{B} + Ax + A\bar{B}$$

$$Q_B^+ = A\bar{B}\bar{x} + AB + B\bar{x}$$

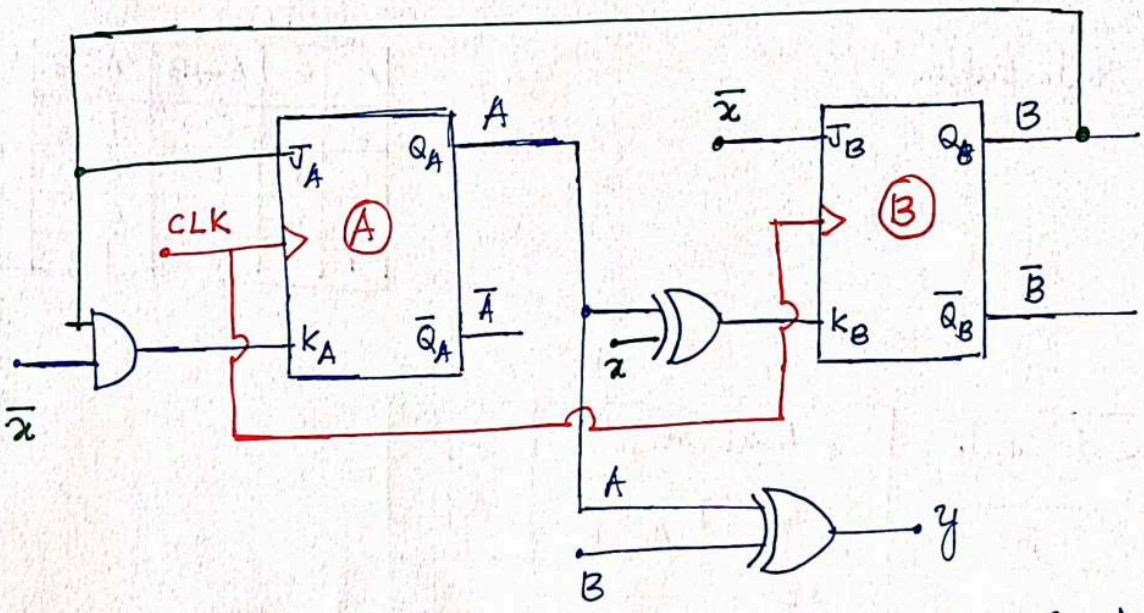
Step 2: State Table

State no.	present state A	present state B	Next state A, x=0	Next state B, x=0	Next state A, x=1	Next state B, x=1	o/p(y) x=0	o/p(y) x=1
a	0	0	0	0 (a)	1	0 (c)	0	0
b	0	1	0	1 (b)	0	0 (a)	0	0
c	1	0	1	1 (d)	1	0 (c)	0	1
d	1	1	0	1 (b)	1	1 (d)	0	0

Step 3: State Diagram



2)



Step 1: Determine the equations for the flip-flop inputs and the equation for the output (y)

$J_A = B$; $J_B = \bar{x}$

$K_A = B\bar{x}$; $K_B = A \oplus x$

We have characteristic / Next state equation of JK FF

$$Q^+ = \bar{Q}J + Q\bar{K}$$

Apply to 'A' FF

$$Q_A^+ = \bar{Q}_A J_A + Q_A \bar{K}_A$$

$$= \bar{A}B + A \cdot \overline{B \cdot \bar{x}}$$

$$= \bar{A}B + A \cdot (\bar{B} + \bar{\bar{x}})$$

$$= \bar{A}B + A\bar{B} + Ax$$

$$\boxed{Q_A^+ = A \oplus B + Ax}$$

Apply to 'B' FF

$$Q_B^+ = \bar{Q}_B J_B + Q_B \bar{K}_B$$

$$= \bar{B}\bar{x} + B\overline{(A \oplus x)}$$

$$\boxed{Q_B^+ = \bar{B}\bar{x} + B(A \odot x)}$$

$$\therefore \boxed{y = A \oplus B}$$

$$Q_A^+ = A \oplus B + Ax \; ; \quad Q_B^+ = \bar{B}\,\bar{x} + B(A \odot x) \; ; \quad y = A \oplus B$$

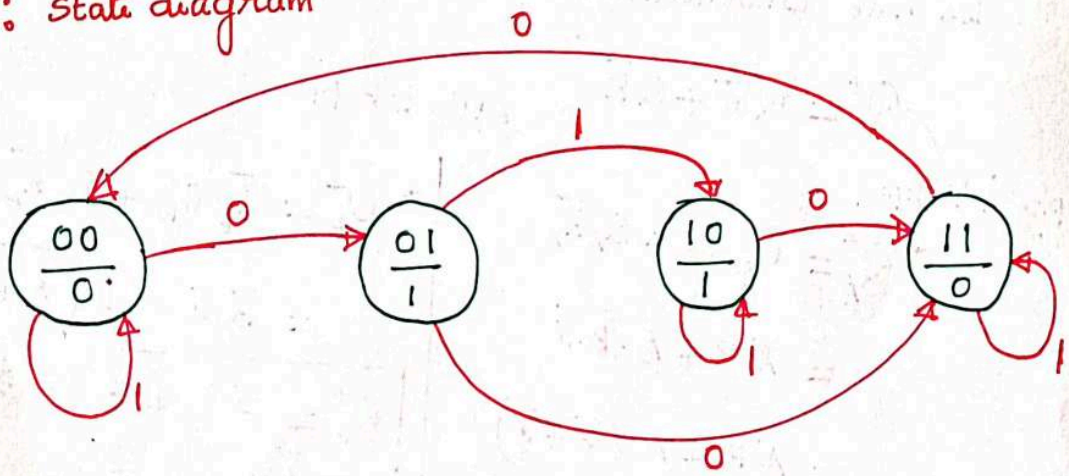
NOTE

A	B	$A \oplus B$	$A \odot B$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	0	1

Step 2 : State table

present state	Next state		o/p (y)
A B	A B	A B	
	$x=0$	$x=1$	
0 0	0 1	0 0	0
0 1	1 1	1 0	1
1 0	1 1	1 0	1
1 1	0 0	1 1	0

Step 3 : State diagram

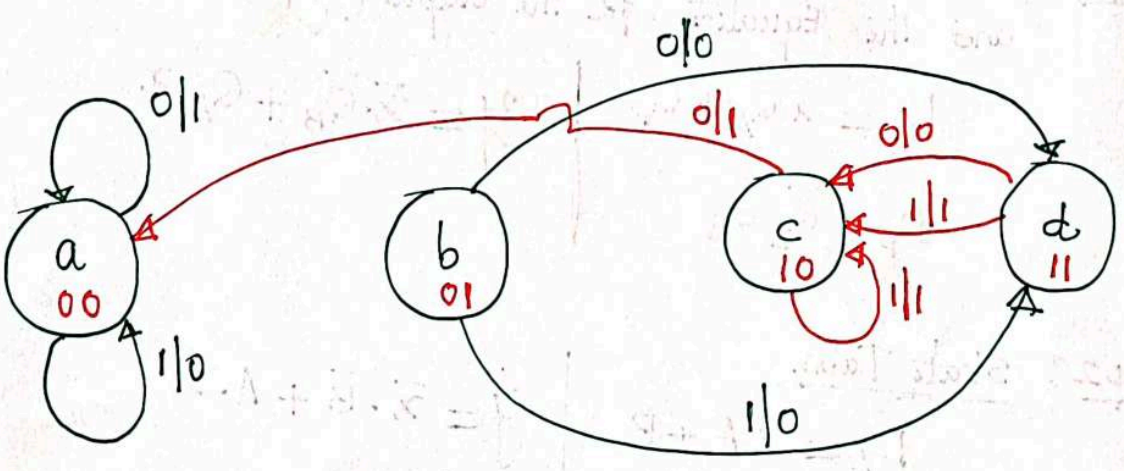


$Q_A^+ = D_A$ | $Q_B^+ = D_B$ | $y = \bar{x}.\bar{B} + A.x$

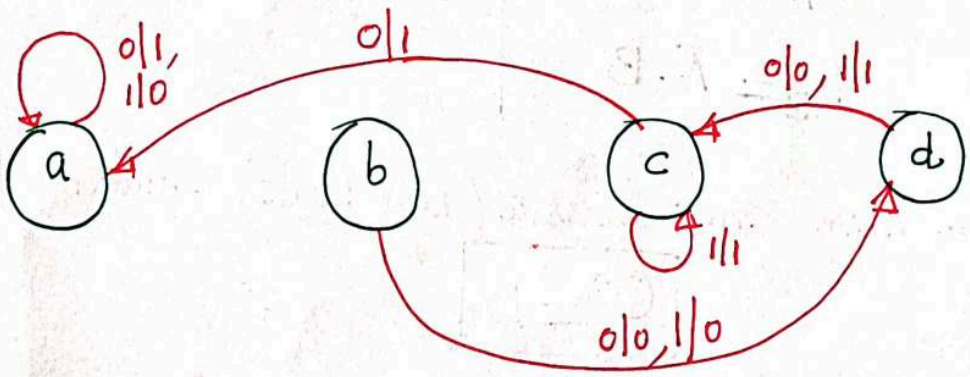
$Q_A^+ = xA + B$ | $Q_B^+ = \bar{A}.B$

	present state A (Q_A)	present state B (Q_B)	Next state A (Q_A^+) x=0	Next state B (Q_B^+) x=0	Next state A (Q_A^+) x=1	Next state B (Q_B^+) x=1	o/p (y) x=0	o/p (y) x=1
a	0	0	0	0	0	0	1	0
b	0	1	1	1	1	1	0	0
c	1	0	0	0	1	0	1	1
d	1	1	1	0	1	0	0	1

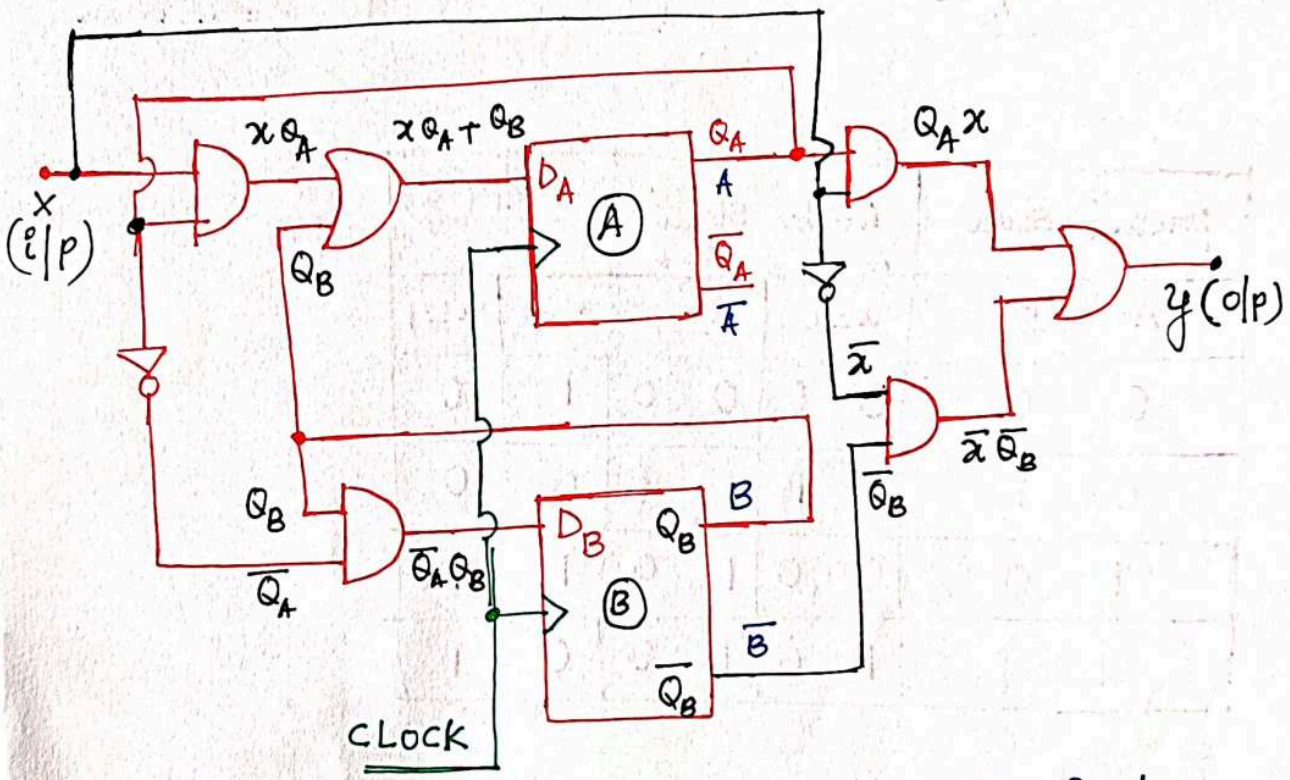
Step 3 : State Diagram



OR



* Analysis of synchronous sequential circuit :



Step1 : Determine the Equations for the flip-flop inputs and the Equation for the output (y)

$$D_A = xQ_A + Q_B$$

$$D_B = \bar{Q}_A Q_B$$

$$y = \bar{x}\cdot\bar{Q}_B + Q_A \cdot x$$

Step2 : State Table

$$D_A = xA + B$$

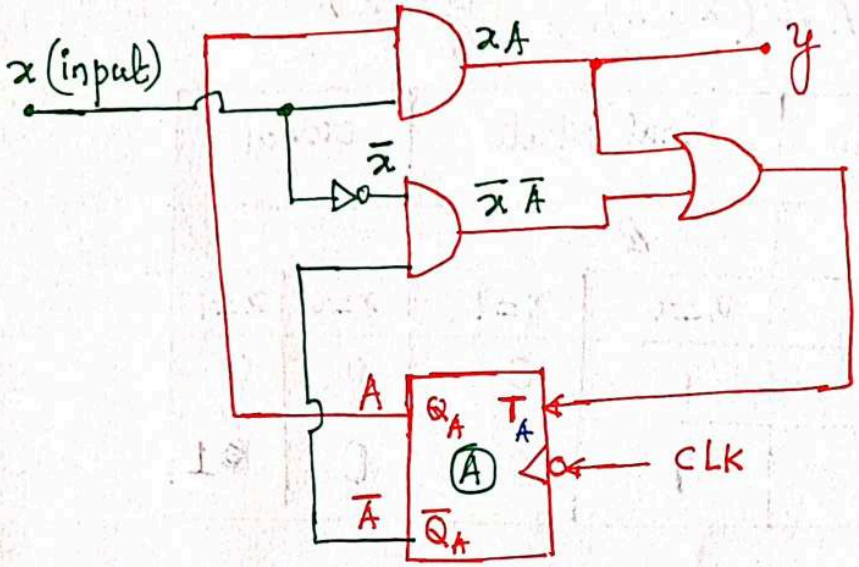
$$D_B = \bar{A}\cdot B$$

$$y = \bar{x}\cdot\bar{B} + A\cdot x$$

We have characteristic (Next state) equation of 'D' FF.

$$\boxed{Q^+ = D}$$

* Analysis of Synchronous Sequential circuit Analysis



Step1 : $T_A = xA + \bar{x}\bar{A}$; $y = xA$

We have characteristic / Next state Equation

$$Q^+ = Q\bar{T} + \bar{Q}T$$

$$Q^+ = Q \oplus T$$

Next state Equation of 'A' FF

$$Q_A^+ = Q_A \oplus T_A$$

$$Q_A^+ = Q_A \oplus (xA + \bar{x}\bar{A}) = Q_A \oplus (x \odot A)$$

$$Q_A^+ = A \oplus (x \odot A)$$

NOTE :

A	B	$A \oplus B$	$A \odot B$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	0	1

Step2 : State Table

$$Q_A^+ = A \oplus (x \odot A) \; ; \quad y = x A$$

	present state (Q_A)	Next state (Q_A^+)		output	
	A				
		x=0	x=1	x=0	x=1
a	0	1	0	0	0
b	1	1	0	0	1

Step3 : State diagram

